

QUANTITATIVE METHODS

RATE AND RETURNS

Required Rate of Return

Interest rate = Real risk-free rate
 + Inflation premium
 + Default risk premium
 + Liquidity premium
 + Maturity premium

Nominal Risk-free Rate

$(1 + \text{Nominal risk-free rate})$
 $= (1 + \text{Real risk-free rate})(1 + \text{Inflation premium})$

Nominal risk-free rate

$\approx \text{Real risk-free rate} + \text{Inflation premium}$

Holding Period Return

$$R = \frac{P_1 - P_0}{P_0} + I$$

$$R = [(1 + R_1)(1 + R_2) \dots (1 + R_N)] - 1$$

Arithmetic Mean Return

$$\bar{R}_i = \frac{\sum_{t=1}^T R_t}{T}$$

Geometric Mean Return

$$R_{GI} = \sqrt[T]{\prod_{t=1}^T (1 + R_{ti})} - 1$$

Harmonic Mean Return (Cost Averaging)

$$\bar{R}_H = \frac{n}{\sum_{i=1}^n \frac{1}{X_i}}, \text{ where } X_i > 0 \text{ for } i = 1, 2, \dots, n$$

If returns are volatile, $\bar{R}_{\text{Arith.}} > \bar{R}_{\text{Geo.}} > \bar{R}_{\text{Har.}}$

$$(\bar{R}_{\text{Arith.}}) \times (\bar{R}_{\text{Har.}}) = (\bar{R}_{\text{Geo.}})^2$$

Money-Weighted Return (MWR)

- IRR derived from all cash inflows and returns
- Can be skewed by timing/value of cash flows
- Appropriate if manager controls timing of CFs

Time-Weighted Return (TWR)

- Geometric mean of sub-period returns
- Compound growth for an initial \$1 investment
- Unaffected by timing/value of cash flows

Future Value (FV) and Present Value (PV)

$$FV_N = PV(1 + r)^N \Leftrightarrow PV = FV_N(1 + r)^{-N}$$

$$FV_N = PV \left(1 + \frac{r_s}{m}\right)^{mN} \Leftrightarrow PV = FV_N \left(1 + \frac{r_s}{m}\right)^{-mN}$$

Continuously Compounded Returns

$$FV_N = PVe^{r_s N}$$

Real Returns

$$(1 + \text{Real return}) = \frac{(1 + r_f)(1 + \text{Risk premium})}{1 + \text{Inflation premium}}$$

Leveraged Return

$$R_L = R_P + \frac{V_B}{V_E}(R_P - r_D)$$

THE TIME VALUE OF MONEY IN FINANCE

Fixed-Income Instruments and the Time Value of Money

Discount instruments/zero-coupon bonds:

- No coupons; investors pay discounted value of bond at inception

Coupon instruments:

$$PV = \frac{PMT_1}{(1 + r)^1} + \frac{PMT_2}{(1 + r)^2} + \dots + \frac{PMT_N + FV_N}{(1 + r)^N}$$

Perpetual bonds/perpetuity:

$$PV = \frac{PMT}{r}$$

Annuity instruments:

$$PV = A \left[\frac{1 - (1 + r)^{-N}}{r} \right]$$

Equity Instruments and the Time Value of Money

Constant dividends:

$$PV = \frac{D}{r}$$

Constant dividend growth rate:

$$PV = \frac{D_1}{r - g} = \frac{D_0(1 + g)}{r - g}$$

Changing dividend growth rate:

$$PV = \sum_{i=1}^n \frac{D_0(1 + g_s)^i}{(1 + r)^i} + \frac{E(S_n)}{(1 + r)^n}$$

$$\text{where } E(S_n) = \frac{D_{n+1}}{r - g_l}$$

Implied Return for Fixed-Income Instruments

Discount bond implied return:

$$r = \left(\frac{FV_N}{PV} \right)^{1/N} - 1$$

Equity Instruments, Implied Return, and Implied Growth

Equity required rate of return:

$$r = \frac{D_1}{PV} + g$$

Forward price-to-earnings ratio:

$$PV/E_1 = \frac{D_1/E_1}{r - g}$$

STATISTICAL MEASURES OF ASSET RETURNS

Measures of Central Tendency

Sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i; n = \text{sample size}$$

Median:

- If n is odd: Median = $\left(\frac{n+1}{2}\right)$ th term

- If n is even:

Median = average of $\left(\frac{n}{2}\right)$ th and $\left(\frac{n+2}{2}\right)$ th terms

Mode:

- The most frequently occurring value

Dealing with Outliers

1. No adjustments
2. Remove all outliers with a trimmed mean
3. Replace outliers with a winsorized mean

Measures of Location

Interquartile range (IQR):

$$IQR = Q_3 - Q_1$$

Location of yth percentile, $L_y = (n + 1) \frac{y}{100}$

If L_y is not an integer, use linear interpolation.

Distributions may be divided into quarters (Quartiles), fifths (Quintiles), or tenths (Deciles)

E.g., 50th percentile = 2nd quartile = 5th decile

Range

Range = Maximum value – Minimum value

Mean Absolute Deviation

$$MAD = \frac{1}{n} \sum_{i=1}^n |X_i - \bar{X}|$$

Sample Variance and Sample Standard Deviation

Sample variance:

$$s^2 = \frac{1}{n - 1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Sample standard deviation is square root of the sample variance

Downside Deviation and Coefficient of Variation

Sample target semideviation:

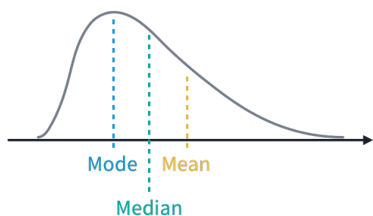
$$s_{\text{Target}} = \sqrt{\frac{\sum_{X_i \leq B} (X_i - B)^2}{n - 1}}$$

Coefficient of variation measures dispersion relative to mean:

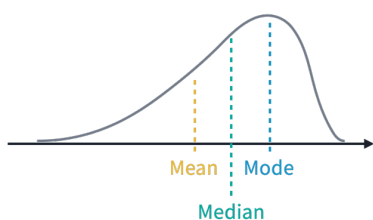
$$CV = \frac{s}{\bar{X}}$$

Skewness

Distribution Skewed to the Right
(Positively Skewed)



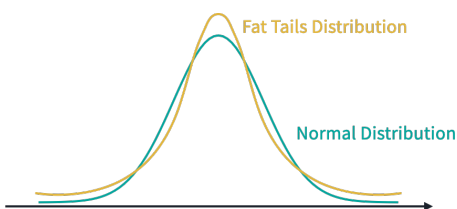
Distribution Skewed to the Left
(Negatively Skewed)



$$\text{Skewness} \approx \left(\frac{1}{n}\right) \frac{\sum_{i=1}^n (X_i - \bar{X})^3}{s^3}$$

Kurtosis (Excess Kurtosis = Kurtosis - 3)

Distribution	Tails	Peaked	Kurtosis
Leptokurtic	Fatter	More	>3
Mesokurtic	Normal	Normal	3
Platykurtic	Thinner	Less	<3



$$\text{Excess kurtosis, } K_E \approx \left(\frac{1}{n}\right) \frac{\sum_{i=1}^n (X_i - \bar{X})^4}{s^4} - 3$$

Sample Covariance

$$s_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Sample Correlation Coefficient

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}$$

PROBABILITY TREES AND CONDITIONAL EXPECTATIONS

Expected Value and Variance

Expected value:

$$E(X) = \sum_{i=1}^n P(X_i) X_i$$

Variance:

$$\sigma^2(X) = \sum_{i=1}^n P(X_i) [X_i - E(X)]^2 = E(X^2) - [E(X)]^2$$

Total Probability Rule for Expected Value

$$E(X) = E(X|S_1)P(S_1) + \dots + E(X|S_n)P(S_n)$$

Conditional Expected Values and Conditional Variances

Conditional expected value:

$$E(X|S) = P(X_1|S)X_1 + \dots + P(X_n|S)X_n$$

Conditional variance:

$$\sigma^2(X|S) = [X_1 - E(X|S)]^2 P(X_1|S) + \dots + [X_n - E(X|S)]^2 P(X_n|S)$$

Bayes' Formula

$$P(\text{Event}|\text{Info}) = \frac{P(\text{Info}|\text{Event})}{P(\text{Info})} \times P(\text{Event})$$

Prior probabilities are updated to become posterior probabilities based on new information

PORTFOLIO MATHEMATICS

Expected Value & Variance of Portfolio Return

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

$$\text{Cov}(R_i, R_j) = E[(R_i - E(R_i))(R_j - E(R_j))]$$

$$\sigma^2(R_p) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(R_i, R_j)$$

An asset's covariance with itself is its variance

Correlation:

$$\rho(R_i, R_j) = \rho_{i,j} = \frac{\text{Cov}(R_i, R_j)}{\sigma(R_i)\sigma(R_j)}$$

For portfolio with 2 investments:

$$E(R_p) = w_A R_A + w_B R_B$$

$$\text{Cov}(R_A, R_B) = \sigma(R_A)\sigma(R_B)\rho(R_A, R_B)$$

$$\sigma^2(R_p) = w_A^2 \sigma^2(R_A) + w_B^2 \sigma^2(R_B) + 2w_A w_B \text{Cov}(R_A, R_B)$$

Forecasting Correlation of Returns

If X and Y are independent:

$$P(XY) = P(X)P(Y)$$

$$E(XY) = E(X)E(Y)$$

Roy's Safety-First Ratio

$$\text{SFRatio} = \frac{E(R_p) - R_L}{\sigma_p}$$

where R_L is the threshold level

$$P(R_p < R_L) = N(-\text{SFRatio})$$

SIMULATION METHODS

Lognormal Distribution

- If Y follows a lognormal distribution, then $\ln(Y)$ follows a normal distribution
- Used to model asset prices
- Positively skewed

Mean and variance of Y:

$$\mu_Y = e^{\mu + 0.5\sigma^2}$$

$$\sigma_Y^2 = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$$

where μ and σ^2 are the mean and variance of $\ln(Y)$

Continuously Compounded Rates of Return

Continuously compounded return from 0 to T:

$$r_{0,T} = \ln\left(\frac{S_T}{S_0}\right)$$

ESTIMATION AND INFERENCE

Sampling Methods

Simple random sampling: Subset of population is chosen at random

Systematic sampling: Every k^{th} observation is chosen until desired sample size is achieved

Stratified sampling: Simple random samples are drawn from each subpopulation (strata)

Cluster sampling: Sample set is divided into mini-representations of the population (cluster)

Convenience sampling: Samples are selected based on accessibility

Judgmental sampling: Samples are selected based on researchers' knowledge and expertise

Sampling error = Sample mean - Population mean

Central Limit Theorem (CLT)

For a sample of size $n \geq 30$ from a population with mean μ and variance σ^2 , the sample mean $\bar{X}$ approximately follows a normal distribution with mean μ and variance σ^2/n

Standard Error of Sample Mean

Population variance is known: $\sigma_{\bar{x}} = \sigma/\sqrt{n}$

Population variance is not known: $s_{\bar{x}} = s/\sqrt{n}$

Bootstrapping and Empirical Sampling Distributions

Bootstrap: Replace each drawn sample with an identical element for the next draw

Jackknife: Draw each sample by leaving out one observation at a time without replacement

HYPOTHESIS TESTING

The Process of Hypothesis Testing

1. State hypotheses (null and alternative)
2. Identify test statistic
3. Specify significance level
4. State decision rule
5. Collect data; calculate test statistic
6. Make decision regarding hypothesis

Hypothesis Test Results

Type	Hypotheses	Reject H_0 if test statistic is
One-tailed (upper)	$H_0: \mu \leq \mu_0$ $H_a: \mu > \mu_0$	> critical value
One-tailed (lower)	$H_0: \mu \geq \mu_0$ $H_a: \mu < \mu_0$	< critical value
Two-tailed	$H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$	< lower critical value or > upper critical value

Hypothesis Testing Decision Errors

Decision	H_0 is True	H_0 is False
Do not reject H_0	Correct	Type II (β)
Reject H_0	Type I (α)	Correct

Power of a test = $1 - P(\text{Type II error}) = 1 - \beta$

p-value: smallest value of α at which H_0 is rejected

Test Concerning a Single Mean

A t-test can be used if the sample:

- Has a large size (from any population with unknown variance); or
- Is approximately normal

$$t\text{-statistic} = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

$$df = n - 1$$

Test Concerning Differences between Means

Normal populations with unknown variances that are assumed equal:

$$t\text{-statistic} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$df = n_1 + n_2 - 2$$

Test Concerning Mean Differences

Normal populations with unknown variances:

$$t\text{-statistic} = \frac{\bar{d} - \mu_{d0}}{s_d}$$

$$df = n - 1$$

Test Concerning a Single Variance

Normal population:

$$\chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}$$

$$s^2 = \frac{1}{n - 1} \sum_{i=1}^n (X_i - \bar{x})^2$$

$$df = n - 1$$

Test Concerning the Equality of Two Variances

Normal populations:

$$F = \frac{s_1^2}{s_2^2}$$

$$df_1 = n_1 - 1$$

$$df_2 = n_2 - 1$$

Nonparametric Tests

Test that is not concerned with parameter and is implemented in situations such as:

- Data do not meet distributional assumptions
- Data are subject to outliers
- Data are given in ranks
- Hypothesis does not concern a parameter

PARAMETRIC AND NON-PARAMETRIC TESTS OF INDEPENDENCE

Tests Concerning Correlation

$$t\text{-statistic} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$df = n - 2$$

To use the Spearman rank correlation coefficient, substitute the following value into the t-statistic calculation:

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Tests of Independence Using Contingency Table Data

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$\text{Standardized residual} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}}}$$

SIMPLE LINEAR REGRESSION

Estimating the Parameters of a Simple Linear Regression

Y: Dependent variable/explained variable

X: Independent variable/explanatory variable

$$Y = b_0 + b_1X + \epsilon$$

where b_0 is the intercept, b_1 is the slope coefficient, and ϵ is the error term

The parameters can be estimated by:

$$\hat{b}_1 = \frac{\text{Cov}[Y, X]}{\text{Var}[X]} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{b}_0 = \bar{Y} - \hat{b}_1\bar{X}$$

$$r = \frac{\text{Cov}[Y, X]}{\sqrt{\text{Var}[Y]}\sqrt{\text{Var}[X]}}$$

Assumptions of the Simple Linear Regression Model

- Linear relationship between X and Y
- Homoscedasticity (i.e., constant variance of residuals)
- Independence between X and Y
- Normality of the residuals

Analysis of Variance

Sum of squares error (SSE):

- Unexplained variation in Y

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Sum of squares regression (SSR):

- Explained variation in Y

$$\text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$$

Sum of squares total (SST):

- Total variation in Y

$$\text{SST} = \text{SSE} + \text{SSR} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Coefficient of determination:

$$R^2 = \frac{\text{SSR}}{\text{SST}}$$

$$= r^2 \text{ (if there is only one independent variable)}$$

F-statistic:

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{\text{SSR}/k}{\text{SSE}/(n - [k + 1])}$$

Standard error of regression:

$$s_e = \sqrt{\text{MSE}} = \sqrt{\frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n - 2}}$$

Hypothesis Testing of Individual Regression Coefficients

To test a hypothesis about the slope:

$$t = \frac{\hat{b}_1 - b_1}{s_{\hat{b}_1}}$$

$$s_{\hat{b}_1} = \frac{s_e}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

To test a hypothesis about the intercept:

$$t = \frac{\hat{b}_0 - b_0}{s_{\hat{b}_0}}$$

$$s_{\hat{b}_0} = \sqrt{\text{MSE} \left(\frac{1}{n} + \frac{(\bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right)}$$

Prediction in the Simple Linear Regression Model

Estimated variance of the prediction error for Y:

$$s_f^2 = s_e^2 \left[1 + \frac{1}{n} + \frac{(X - \bar{X})^2}{(n - 1)s_x^2} \right]$$

Functional Forms for Simple Linear Regression

Common functional forms:

- Log-lin model
- Lin-log model
- Log-log model

INTRODUCTION TO BIG DATA TECHNIQUES

Fintech and Big Data

Characteristics of big data:

- Volume
- Velocity
- Variety

Sources of alternative data:

- Individuals (e.g., unstructured data)
- Business processes (e.g., corporate information)
- Sensors (e.g., smartphones and satellite data)

Types of Machine Learning

- Supervised learning
- Unsupervised learning
- Deep learning

ECONOMICS

FIRMS AND MARKET STRUCTURES

Profit maximization: $MR = MC$

Economic profit:

Revenue > Explicit costs + Implicit costs

Normal profit: Economic profit = 0

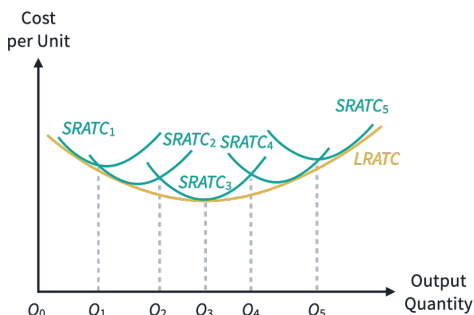


Shutdown Decision (Short-term vs. Long-term)

	Short-Term	Long-Term
$TR \geq TC$	Stay	Stay
$TVC < TR < TC$	Stay	Exit market
$TR < TVC$	Shut down	Exit market

Economies of Scale

Each stage of expansion has its own short-run ATC curve. *Minimum efficient scale* is the low point on the long-run average total cost curve.

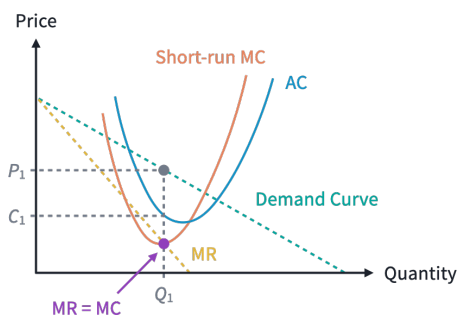


Perfect Competition

- Firms: Many
- Products: Homogeneous
- Barriers to entry: Very low
- Pricing power of firms: None
- Non-price competition: None

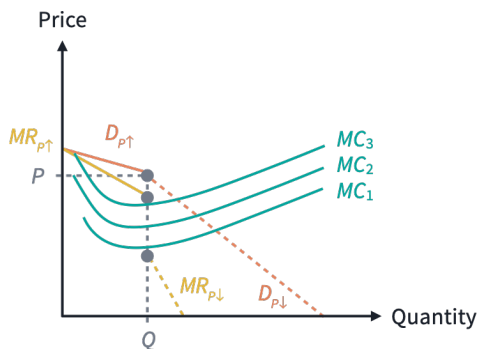
Monopolistic Competition

- Firms: Many
- Products: Differentiated
- Barriers to entry: Low
- Pricing power of firms: Some
- Non-price competition: Advertising and product differentiation



Oligopoly

- Firms: Few
- Products: Homogeneous
- Barriers to entry: High
- Pricing power: Some or considerable
- Non-price competition: Advertising and product differentiation



Kinked demand curve: A price increase will impact sales more than an equivalent price decrease

Cournot assumption: Competitors will maintain current output levels if one firm changes its price

Game theory: If one firm changes its prices, competitors will adjust to maximize their profits, resulting in a Nash equilibrium

Price collusion is more likely to happen if:

- Few firms or one dominant firm
- Products are relatively similar
- Firms have similar cost structures
- Orders are frequent and relatively small
- Credible threat of retaliation for breaking pact
- The threat of external competition is high

Monopoly

- Firm: One
- Products: Unique
- Barriers to entry: Very high
- Pricing power of firm: Considerable
- Non-price competition: Advertising

Market Power Measures

N-firm concentration ratio: Sum of market share of the N largest firms in the industry

Herfindahl-Hirschman Index (HHI): Sum of squared market share of the N largest firms

UNDERSTANDING BUSINESS CYCLES

Business Cycle Phases

Recovery

- *Economy:* Going through a trough
- *Activity level:* Below potential but start to increase
- *Employment:* Layoffs slow, but firms prefer extending overtime to rehiring full-time
- *Inflation:* Moderate
- *Capital spending:* Low but increasing, with a focus on efficiency rather than capacity

Expansion

- *Economy:* Enjoying an upswing
- *Activity level:* Above-average growth rates
- *Employment:* Full-time rehiring, more overtime
- *Inflation:* Moderate, but increasing
- *Capital spending:* Focused on capacity expansion

Slowdown

- *Economy:* Going through a peak
- *Activity level:* Decelerating
- *Employment:* Hiring slows
- *Inflation:* Accelerating
- *Capital spending:* Strong capital spending, but inventory starts building up as sales growth slows

Contraction

- *Economy:* Weakens and may go into a recession
- *Activity levels:* Below potential
- *Employment:* Hiring freezes, then layoffs
- *Inflation:* Decelerating, but with a lag
- *Capital spending:* New orders halted and existing orders canceled, scale back on maintenance

Economic Indicators

Leading: Stock indexes, building permits

Coincident: Real income, industrial production

Lagging: Unemployment rate, prime lending rate

FISCAL POLICY

Fiscal Policy: Spending Tools

Transfer payments: Redistribution of wealth (e.g., unemployment benefits)

Current spending: Spending on goods and services

Capital spending: Spending on infrastructure

Fiscal Policy: Revenue Tools

Direct taxes: Tax on income (e.g., income taxes, corporate taxes, capital gains taxes)

Indirect taxes: Tax on goods and services

Fiscal Multiplier

$$= \frac{1}{1 - MPC(1 - t)}$$
, where MPC = marginal propensity to consume; t = tax rate

Difficulties Executing Fiscal Policy

Recognition lag: Government must see need

Action lag: Time needed to choose policy

Impact lag: Policies do not have immediate impact

MONETARY POLICY

Roles of Central Banks

- Supplier of currency
- Banker to government and bankers' bank
- Lender of last resort
- Regulator of payments system
- Conductor of monetary policy
- Supervisor of banking system
- Maintain foreign currency reserves and gold reserves

Monetary Policy Tools

Expansionary monetary policy measures:

- **Open market operations:** Buy bonds from commercial banks
- **Policy rate:** Set policy rate below neutral level
- **Reserve requirement:** Reduce reserves for commercial banks

INTRODUCTION TO GEOPOLITICS

National Governments and Political Cooperation

State actors possess the authority to deploy a country's national security resources

Non-State Actors and the Forces of Globalization

Non-state actors participate in global political, economic, or financial affairs but do not control a country's national security resources

Assessing Geopolitical Actors and Risk



The Tools of Geopolitics

National security tool: Military force, espionage

Economic tools: Currency union, nationalization

Financial tools: Currency markets, sanctions, capital controls

Incorporating Geopolitical Risk into the Investment Process

Types of geopolitical risk:

- Event risk
- Exogenous risk
- Thematic risk

Assessing Geopolitical Threats

To assess geopolitical risk, consider:

- The likelihood of occurrence
- The velocity (speed) of impact
- The size and nature of impact

INTERNATIONAL TRADE AND CAPITAL FLOWS

Benefits and Costs of International Trade

Benefits:

- Gains from exchange
- Greater economies of scale
- Greater product variety
- Increased competition
- More efficient allocation of resources

Costs:

- Income inequality
- Job losses

Trade Restrictions

Tariffs: Taxes on imported goods

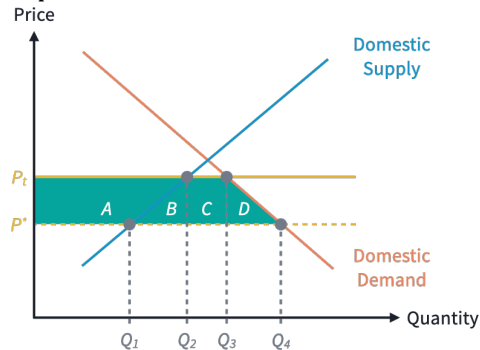
Quotas: Limits on quantity of imported goods

Export subsidies: Payments to exporters

Minimum domestic content requirements

Voluntary export restraints: Self-imposed limitations by foreign producers

Impact of Trade Restrictions



- Price increases from P^* to P_t
- Domestic production increases from Q_1 to Q_2
- Domestic consumption falls from Q_4 to Q_3
- Imports fall from $(Q_4 - Q_1)$ to $(Q_3 - Q_2)$
- Loss of consumer surplus = $(A + B + C + D)$
- National welfare loss = $(B + D)$
- Increase in producer surplus = A
- Tariff revenue/Quota rent = C

Regional Trading Blocs

Free trade area (FTA): Free trade among members

Customs union (CU): FTA + common trade policy

Common market (CM): CU + free movement of factors of production within bloc

Economic union (EU): CM + common economic institutions and coordination of economic policies

Monetary union (MU): EU + common currency

CAPITAL FLOWS AND THE FX MARKET

The Foreign Exchange Market and Exchange Rates

$$\text{Real ex. rate}_{d/f} = \text{Nominal ex. rate}_{d/f} \times \left(\frac{\text{CPI}_f}{\text{CPI}_d} \right)$$

"A/B": How many units of currency A can be purchased by one unit of currency B

Ideal Currency Regime

1. Exchange rates are credibly fixed
2. Fully convertible currencies, free capital flows
3. Countries pursue independent monetary policies

Such an ideal currency regime is NOT possible

Exchange Rate Regimes

Dollarization: Adopt another country's currency

Monetary union: Adopt a common currency

Currency board: Commitment to exchange domestic currency at fixed exchange rate

Fixed peg: Currency is pegged to foreign currency (or basket of currencies) within $\pm 1\%$ margin

Target zone: Fixed peg with wider margin

Crawling peg: Peg rate is periodically adjusted

Crawling bands: Margin increases over time, usually to transition from fixed peg to floating

Managed floating: Monetary authority intervenes, but no official target exchange rate

Independently floating: Market sets exchange rate

Trade Balance

$$X - M = (S - I) + (T - G)$$

A trade surplus must be reflected in excess private savings or fiscal surplus

EXCHANGE RATE CALCULATIONS

Cross-Rate Calculations

$$S_{A/B} = S_{A/C} \times S_{C/B}$$

Forward Rate Calculations

$$\frac{\text{Forward exchange rate}_{d/f}}{\text{Spot exchange rate}_{d/f}} = \frac{1 + i_d}{1 + i_f}$$

Forward exchange rates in points:

- Unit of points is last decimal place in the rate quote (e.g., 1.5301 to 1.5302 is a 1-point increase)

CORPORATE ISSUERS

ORGANIZATIONAL FORMS, CORPORATE ISSUER FEATURES, AND OWNERSHIP

Sole Proprietorship

- Extension of owner
- Operated by owner
- Business liability is retained by owner
- Business profits are owned by owner and taxed as personal income
- Owner is the main source of capital

General Partnership

- Set by partnership agreement
- Operated by partners
- Business liability is retained and shared by partners
- Business profits are shared by partners and taxed as personal income
- Partners are the main source of capital
- Partners' resources and risk appetite limit business growth

Limited Partnership

- Set by partnership agreement
- Operated by GP
- Business liability is limited for LPs and unlimited for GP
- Business profits are shared by partners and taxed as personal income
- Partners' resources, risk appetite, and GP's competence/integrity limit business growth

Limited Liability Partnership

- Set by partnership agreement
- Operated by LPs and there is no GP
- Business liability is limited for LPs
- Business profits are shared by partners and taxed as personal income

Limited Companies

- Legal identity is separated from owners
- Operated by management team voted by shareholders
- Limited business liability for shareholders
- Financed by equity and debt



Private limited companies:

- Profits are taxed at the personal level
- Subject to restrictions on the number of owners
- Votes are needed to transfer ownership interest

Public limited companies:

- Double taxation: once at the corporate level and once at the personal level (for dividend income)
- No restrictions on the number of owners
- Ownership interest can be transferred anytime

Publicly vs. Privately Owned Corporate Issuers

Private placement memorandum (PPM) is used by private companies to raise capital in primary market

Private companies can go public by:

- Initial public offering (IPO)
- Direct listing (DL)
- Acquisition

Public companies can go private by:

- Leveraged buyout (LBO)

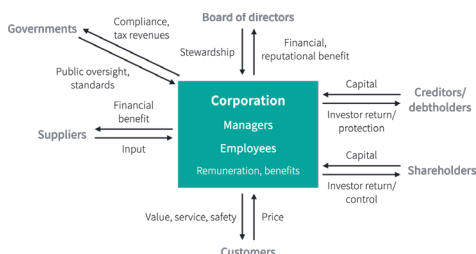
INVESTORS AND OTHER STAKEHOLDERS

Financial Claims of Lenders and Shareholders

Risk vs return characteristics of equity and debt:

	Equity	Debt
Upside potential	Unlimited	Limited to payments
Maximum loss	Cannot be more than the investment value	
Investment risk	Higher	Lower
Investment interest	Maximize company value	Timely repayment

Corporate Stakeholders and Governance



CORPORATE GOVERNANCE: CONFLICTS, MECHANISMS, RISKS, AND BENEFITS

Stakeholder Conflicts and Management

Shareholder vs. manager/director

- *Entrenchment*: Managers avoid justifiable risks to avoid losing their positions
- *Empire building*: Making unjustified acquisitions to increase company size and compensation
- *Excessive risk taking*: Taking unjustifiable risks to maximize returns on stock-based compensation
- *Agency costs* reduce the potential for exploitation in an agency relationship

Controlling shareholder vs. minority shareholder

- *Dispersed ownership*: Controlled by many minority shareholders
- *Concentrated ownership*: Controlled by a single shareholder
- *Multiple-class share structures*: Disproportionate voting power to certain shareholder classes

Shareholder vs. creditor

- Equity owners prefer growth and have a higher risk tolerance
- Creditors prefer stability and limited downside risk

Corporate Governance Mechanisms

Shareholder:

- Corporate reporting and transparency
- Shareholder meetings (cumulative voting, proxy voting)
- Shareholder activism
- Derivative lawsuits
- Corporate takeovers (proxy contests, tender offers, hostile takeovers)

Creditor:

- Bond indentures, collateral, and trustees
- Corporate reporting
- Creditor committees

Board and management:

- Audit committee
- Governance committee
- Remuneration committee
- Nomination committee
- Risk committee
- Investment committee

Employee:

- Labor laws
- Code of ethics and compliance department
- Whistleblower protections
- Employee contracts

Customer and supplier:

- Commercial contracts
- Public reputation and social media

Government:

- Laws and regulations
- Corporate governance codes
- Common law and civil law systems

Corporate Governance Risks and Benefits

Operational risks of poor stakeholder governance:

- Weak control systems that do not treat all stakeholders fairly
- Ineffective decision-making process
- Inadequate board scrutiny
- Diminished operating performance

Financial risks of poor stakeholder governance:

- Higher default and bankruptcy risks
- Higher borrowing costs
- Poor equity returns

Factors relevant to corporate governance and stakeholder management analysis:

- Economic ownership and voting control
- Board of directors representation
- Remuneration and company performance
- Investors in the company
- Strength of shareholders' rights
- Management of long-term risks

WORKING CAPITAL AND LIQUIDITY

Cash Conversion Cycle

Cash conversion cycle = DOH + DSO - DPO

Trade credit:

EAR of supplier financing

$$= \left[\left(1 + \frac{d}{1-d} \right)^{\frac{365}{\text{Payment period} - \text{Discount period}}} \right] - 1$$

where d is the discount percentage

Working capital = Current assets

- Current liabilities

Liquidity

Primary sources of liquidity:

- Free cash flows
- Ready cash balances (bank accounts)
- Short-term funds (lines of credit)
- Cash flow management (centralized collection)

CFO

= Cash received from customers

+ Interest and dividends from investments

- Cash paid to employees

- Cash paid to suppliers

- Cash paid to government for tax obligations

- Cash paid to lenders for interest obligations

Free cash flow (to equity)

= CFO - Investments in long term assets

Secondary sources of liquidity:

- Negotiating debt contracts
- Liquidating assets
- Filing for bankruptcy

Drag on liquidity: Delayed cash inflows, such as uncollected receivables and obsolete inventory

Pull on liquidity: Accelerated cash outflows, such as settling payables earlier

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Cash} + \frac{\text{Marketable securities}}{\text{securities}} + \text{Receivables}}{\text{Current liabilities}}$$

$$\text{Cash ratio} = \frac{\text{Cash} + \text{Marketable securities}}{\text{Current liabilities}}$$

Working Capital Management

Conservative approach:

Advantages:

- Low rollover risk
- Greater cash flow certainty
- Low risk of inventory shortages
- Flexibility to adapt to adverse market conditions

Disadvantages:

- High borrowing costs
- High cost of equity and shareholder dilution
- Less flexibility to borrow on an as-needed basis
- Longer lead times
- More covenants
- High risk of obsolete inventory

Aggressive approach:

Advantages:

- Low financing costs under an upward-sloping yield curve
- Great flexibility to borrow only as needed
- Short-term borrowing involves less rigorous credit analysis

Disadvantages:

- Risk of having to refinance at higher short-term rates
- Potential difficulty rolling over short-term debt with market turmoil
- Possible need to rely on expensive trade credit
- Tight customer credit terms

Liquidity and Short-Term Funding Needs

Factors influencing a company's short-term borrowing strategy:

- Size and creditworthiness
- Legal and regulatory considerations
- Sufficient access
- Flexibility of borrowing options

CAPITAL INVESTMENTS AND CAPITAL ALLOCATION

Types of Capital Investments

Business maintenance:

- Going concern projects
- Regulatory/compliance projects

Business growth:

- Expansion projects
- Pet projects/high-risk investments

Net Present Value (NPV)

Sum of present values of expected future cash inflows, net of initial cash outlay

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1+r)^t} - \text{Outlay}$$

Accept a project if $NPV > 0$

Internal Rate of Return (IRR)

IRR is r such that $NPV = 0$

Accept a project if its IRR $>$ required return

BA II Plus NPV Worksheet Function

- Cash inflows are positive; outflows are negative
- F01, F02, etc. refer to cash flow frequencies
- CPT + NPV to compute NPV; CPT + IRR for IRR

Return on Invested Capital

Return on invested capital (ROIC):

$$ROIC = \frac{\text{After-Tax Net Profit}}{\text{Average BV of Invested Capital}} = \frac{(1 - \text{Tax Rate}) \times \text{Operating Profit}}{\text{Average (LT Liabilities + Equity)}}$$

Capital Allocation Pitfalls

- Internal forecasting errors
- Ignoring costs of internal financing
- Inconsistent treatment of inflation
- Inertia
- Basing decisions on earnings metrics
- Pet projects
- Failing to consider alternatives

Real Options

- *Timing option*: Option to delay the investment
- *Sizing option*: Option to expand, grow, or abandon
- *Flexibility option*: Option to alter operations, such as changing prices or substituting inputs
- *Fundamental option*: Option to alter decisions based on future events (e.g., drill based on price of oil, continue R&D depending on initial results)

Analyzing Projects with Real Options

- Use the discounted cash flow (DCF) analysis without considering real options
- Adjust the stand-alone DCF analysis by including the present value of the expected costs and benefits options
- Use option pricing models
- Use decision trees

CAPITAL STRUCTURE

Weighted Average Cost of Capital (WACC)

$$WACC = w_d r_d (1 - t) + w_p r_p + w_e r_e$$

w_d = percentage of debt in capital structure

w_p = percentage of preferred stock

w_e = percentage of common stock

t = tax rate

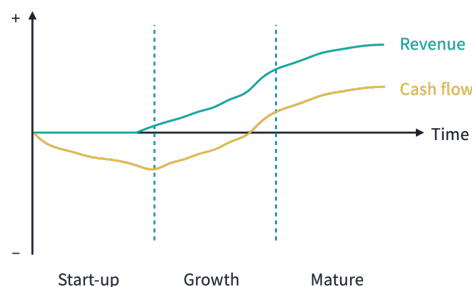
r_d = cost of debt

r_p = cost of preferred stock

r_e = cost of equity

Determinants of the Amount and Type of Financing Needed

Company life cycle:



Determinants of the Costs of Debt and Equity

Issuer-specific factors:

- Sales risk
- Profitability risk: operating leverage
- Financial leverage and interest coverage
- Asset type (tangible, fungible, and liquid)

Modigliani and Miller Propositions

MM Proposition I: A firm's capital structure would have no effect on its value, assuming:

1. Investors have homogeneous expectations
2. No market frictions (e.g., transaction costs, taxes, or costs of financial distress)
3. No agency costs
4. Investors can borrow and lend at risk-free rate
5. Investing/financing decisions are independent

MM Proposition II: Cost of equity increases with the debt-to-equity ratio.

Without Taxes

Firm value	$V_L = V_U$
Cost of Equity	$r_e = r_0 + (r_0 - r_d) \left(\frac{D}{E} \right)$

With Taxes

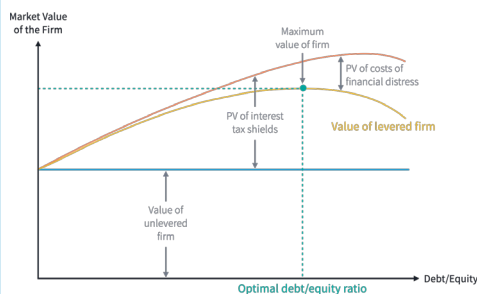
Firm value	$V_L = V_U + tD$
Cost of Equity	$r_e = r_0 + (r_0 - r_d)(1 - t) \left(\frac{D}{E} \right)$

r_0 = cost of capital for a firm financed only by equity

Optimal and Target Capital Structure

Static trade-off theory balances costs of financial distress with tax shield benefits from using debt:

$$V_L = V_U + tD - \text{PV}(\text{costs of financial distress})$$



To estimate a company's target capital structure:

- Assume the company will main its current capital structure
- Infer target weights the company is moving toward
- Use the industrial average

Pecking order theory: Since managers have an asymmetric information advantage, they prefer capital sources that reveal the least amount of information:

1. Internally generated earnings (best option)
2. New debt
3. New equity (least attractive for managers)

BUSINESS MODELS

Business Model Features

- Target customers
- Product/service offering
- Channel strategy
 - Traditional channel
 - Direct sales
 - Drop shipping
 - Omnichannel strategy
- Pricing strategy

Pricing and Revenue Models

Price discrimination:

- Tiered pricing
- Dynamic pricing
- Value-based pricing
- Auction/reverse auction models

Pricing for multiple products:

- Bundling
- Razors-and-blades pricing
- Add-on pricing

Pricing for rapid growth:

- Penetration pricing
- Freemium pricing
- Hidden revenue business model

Alternatives to ownership:

- Subscription pricing
- Leasing
- Licensing
- Franchising
- Fractionalization

Value Chain

Value chain: Systems and functions within the firm that create value for its customers

Supply chain: Series of steps and processes needed to prepare a product to be sold to the consumer

Unit economics: The quantitative analysis of a company's revenues and costs on a per unit basis

Business Model Types

- Private label manufacturers
- Licensing arrangements
- Value added resellers
- Franchise models
- Network effects
- Crowdsourcing

FINANCIAL STATEMENT ANALYSIS

INTRODUCTION TO FINANCIAL STATEMENT ANALYSIS

Regulated Sources of Information

IOSCO: International body of regulatory authorities

SEC: US capital markets regulator

Auditor's reports:

- Unqualified
- Qualified
- Adverse
- Disclaimer

ANALYZING INCOME STATEMENTS

Revenue Recognition

Revenue must not be recognized unless:

- Risks of ownership have been transferred
- Amount of revenue can be reliably measured
- Customer is likely to pay
- Transaction is unlikely to be reversed

Service revenue may be recognized as earned

Allowance for doubtful accounts: Contra-asset account, estimated based on historical experience

Expense Recognition

Matching principle: Expenses must be recognized in the same period as associated revenue

Capitalizing vs. expensing:

- Capitalizing increases assets on the balance sheet and investing cash outflows
- Expensing reduces net income by the after-tax expenditure amount in the period it is incurred

Income Statement Line Items

Revenue

– Cost of goods sold (COGS)

Gross Profit

– Selling, General & Admin. (SG&A)

EBITDA

– Depreciation and Amortization

EBIT (Operating profit)

– Interest

EBT (Earnings before taxes)

– Taxes

Net Income (NI)

Separately Reported Items

- Unusual or infrequent items (US GAAP only)
- Discontinued operations
- Non-operating items

Basic Earnings per Share

Net income – Preferred dividends

Weighted average of shares outstanding

Diluted Earnings per Share

$$\frac{\text{Net income} - \text{Preferred dividends} + \frac{\text{Convertible preferred dividends}}{\text{Shares from preferred shares}} + \frac{\text{Convertible debt interest}}{\text{Shares from convertible debt}}}{\text{Weighted average shares} + \frac{\text{Shares from preferred shares}}{\text{Shares from convertible debt}} + \frac{\text{Convertible debt}}{\text{Shares from convertible debt}} + \frac{\text{Convertible debt interest}}{\text{Shares from convertible debt}}} (1 - t)$$

Diluted EPS must be equal to or less than basic EPS

ANALYZING BALANCE SHEETS

Non-Current Assets

- Intangible assets
- Goodwill
- Financial instruments

Non-Current Liabilities

- Long-term financial liabilities
- Deferred tax liabilities

ANALYZING STATEMENTS OF CASH FLOWS

Relationship between Financial Statements

Ending AP

= Beginning AP

+ (Ending inventory – Beginning inventory)

– Cash paid to suppliers + COGS

Ending inventory

= Beginning inventory

+ (Ending AP – Beginning AP)

+ Cash paid to suppliers – COGS

$\Delta AP + \text{Cash paid to suppliers}$

$= \Delta \text{Inventory} + \text{COGS}$

CFO Direct Method

- Convert each accrual-based item in the income statement to cash inflow/outflow
- CFO is net of cash inflows and outflows

CFO Indirect Method

- Start with net income
- Add noncash expenses (e.g., Depreciation)
- Subtract gains/add losses
- Add increases in current liabilities
- Subtract increases in (non-cash) current assets

Beginning accounts receivable

+ Revenue

– Ending accounts receivable

Cash collected from customers

Cost of goods sold

+ Increase in inventory

Purchases from suppliers

– Increase in accounts payable

Cash paid to suppliers

Differences Between IFRS and US GAAP

Item	IFRS	US GAAP
Interest received	Operating or investing	Operating
Interest paid	Operating or financing	Operating
Dividends received	Operating or investing	Operating
Dividends paid	Operating or financing	Financing
Bank overdrafts	Part of cash equivalents	Financing

Free Cash Flow (FCF)

Free cash flow to the firm (FCFF): Cash available to equity owners and debt holders.

$$\begin{aligned}\text{FCFF} &= \text{NI} + \text{NCC} + \text{I} \times (1 - t) - \text{FCI} - \text{WCI} \\ &= \text{CFO} + \text{I} \times (1 - t) - \text{FCI}\end{aligned}$$

Free cash flow to equity (FCFE): Cash flow available to common shareholders

$$\text{FCFE} = \text{CFO} - \text{FCI} + \text{Net Borrowing}$$

ANALYSIS OF INVENTORIES

Inventory Valuation

IFRS: Lower of cost or net realizable value

US GAAP: Lower of cost or market value

Reversals of inventory write-downs are allowed under IFRS, but not under US GAAP

Inventory Valuation Methods and Systems

	US GAAP	IFRS
FIFO	Allowed	Allowed
LIFO	Allowed	N/A
Weighted average	Allowed	Allowed
Specific Identification	Allowed	Allowed

Impact of Inventory Valuation Method

If prices are rising	FIFO	LIFO
Ending Inventory	Higher	Lower
COGS	Lower	Higher
Net income	Higher	Lower
Income Tax Expense	Higher	Lower
Operating cash flow	Lower	Higher

ANALYSIS OF LONG-TERM ASSETS

Impairment of Property, Plant, and Equipment

Property, plant, and equipment (PP&E):

IFRS

- Both cost model and revaluation model allowed
- Impairment loss
 - = Carrying amount – Recoverable amount
- Recoverable amount is greater of:
 - (1) fair value less selling costs, and
 - (2) value in use (PV of asset's future cash flows)
- Loss recoveries are allowed

US GAAP

- Only cost model is allowed
- Loss recoveries not allowed
- Impairment loss
 - = Carrying amount – Fair value

Impairment of PP&E and Intangible Assets

US GAAP

- Asset tested for impairment only when firm may not recover carrying value through future use
- Asset is impaired when carrying value exceeds asset's future undiscounted cash flows
- Impaired asset's value is written down to fair value and a loss is recognized and cannot be subsequently reversed

IFRS

- Assets are tested annually for impairment
- Impaired if carrying value > recoverable amount
- Impaired asset's value is written down to recoverable amount and a loss is recognized
- Loss can be reversed if asset value recovers, but only up to pre-impairment carrying value

Derecognition

Long-lived assets are derecognized if:

- (1) the asset is disposed of
- (2) there are no expected future benefits from use or disposal

TOPICS IN LONG-TERM LIABILITIES AND EQUITY

Lease Classification

Conditions requiring a lease to be a finance lease:

- Ownership of the leased asset is transferred to the lessee
- Lessee has the option to purchase the asset and will likely do so
- Lease term covers most of asset's useful life
- The present value of lease payments at inception is close to the asset's fair value
- The leased asset is so specialized that only the lessee can use it without modification

Lessee Accounting

US GAAP

Finance lease:

- Lessee purchases the asset, financed by the lessor
- Lessee's periodic lease payments have separate depreciation and interest components

Operating lease (like a rental agreement):

- Single lease expense, not separated into different components for depreciation and interest
- The value of an operating lease payment is calculated as a straight-line allocation of total payments over the term of the lease

IFRS require all leases to be treated in the manner that is prescribed by US GAAP for finance leases.

Lessor Accounting

- For operating leases (under both IFRS and US GAAP), the lessor retains the leased asset on its balance sheet and incurs the associated depreciation expense. Lease income from the lessee is recorded as revenue.
- For finance leases (under both IFRS and US GAAP), the lessor removes the leased asset from its balance sheet and creates an asset with a value equal to the lease receivable and any residual value.
- Lease payments are recognized as an operating inflow on the lessor's cash flow statement (for both operating leases and finance leases)

Pension Plans

Defined benefit (DB): Firm makes periodic payments to employee after retirement.

Components of the change in the net pension asset or liability under IFRS:

- (1) Employees' service cost (income statement)
- (2) Net int. expense/income (income statement)
- (3) Remeasurement (OCI)

Components of the change in the net pension asset or liability under US GAAP:

- (1) Current service cost (income statement)
- (2) Interest expense (income statement)
- (3) Expected return on plan assets (income statement)
- (4) Past service cost (OCI; amortized in income statement)
- (5) Actuarial gains/losses (income statement; amortized in OCI)

Share-Based Compensation

- Stock grants (outright or restricted)
- Stock options
- Stock appreciation rights (SARs)
- Phantom shares

ANALYSIS OF INCOME TAXES

Temporary Differences

Deferred tax liabilities (DTL): Created when taxes payable is less than income tax expense

Deferred tax assets (DTA): Created when taxes payable exceeds income tax expense

Tax base of assets: Amount that will be deducted on the tax return as asset's benefits are realized

Tax base of liabilities: Carrying value of liability minus amount that will be deductible

Asset carrying amount > Tax base	DTL
Asset carrying amount < Tax base	DTA
Liability carrying amount > Tax base	DTA
Liability carrying amount < Tax base	DTL

Impact of tax rate changes

If tax rate increases, DTA and DTL will increase

If tax rate decreases, DTA and DTL will decrease

Valuation Allowance

Contra account used if it is unlikely that future profits will be sufficient to use DTAs and credits

Corporate Income Tax Rates

Statutory tax rate: Corporate income tax rate in domestic market

Effective tax rate: Reported income tax expense as a share of EBT

Cash tax rate: Actual income tax expense as a share of EBT



FINANCIAL REPORTING QUALITY

Spectrum of Financial Reporting Quality

- (1) GAAP, decision-useful, sustainable, and adequate returns
- (2) GAAP, decision-useful, but sustainable?
- (3) Biased accounting choices
- (4) Within GAAP, but "earnings management"
- (5) Departures from GAAP

Depreciation Methods

$$\text{Straight-line:} = \frac{\text{Cost} - \text{Salvage value}}{\text{Depreciable life}}$$

Double-declining balance (DDB):

$$\text{Depreciation}_t = \left(\frac{\text{Book value}_t}{\text{Depreciable life}} \right) \times 2$$

Units-of-production:

$$\text{Depreciation}_t = \frac{\text{Cost} - \text{Salvage}}{\text{Total output}} \times \text{Output units}_t$$

FINANCIAL ANALYSIS TECHNIQUES

Common-Size Analysis

Vertical:

- State income statement items as % of revenue
- State balance sheet items as a % of total assets
- State each cash flow statement item as a % of total cash inflows/outflows

Horizontal (Trend) Analysis:

- State each item relative to its base-year value

Activity Ratios

$$\text{Total asset turnover} = \frac{\text{Revenue}}{\text{Average total assets}}$$

$$\text{Fixed asset turnover} = \frac{\text{Revenue}}{\text{Average net fixed assets}}$$

$$\text{Working capital turnover} = \frac{\text{Revenue}}{\text{Average working capital}}$$

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Average inventory}}$$

$$\text{Receivables turnover} = \frac{\text{Annual sales}}{\text{Average receivables}}$$

$$\text{Payables turnover} = \frac{\text{Purchases}}{\text{Average trade payables}}$$

$$\text{Days of inventory on hand} = \frac{365}{\text{Inventory turnover}}$$

$$\text{Days of sales outstanding} = \frac{365}{\text{Receivables turnover}}$$

$$\text{Number of days of payables} = \frac{365}{\text{Payables turnover}}$$

Liquidity Ratios

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Cash} + \frac{\text{Marketable securities}}{\text{Current liabilities}} + \text{Receivables}}{\text{Current liabilities}}$$

$$\text{Cash ratio} = \frac{\text{Cash} + \text{Marketable securities}}{\text{Current liabilities}}$$

$$\text{Defensive interval} = \frac{\text{Cash} + \text{Marketable securities} + \text{Receivables}}{\text{Average daily expenditures}}$$

$$\text{Cash conversion cycle} = \frac{\text{Days of sales outstanding} + \text{Days of inventory on hand} - \text{Number of days payables}}{\text{payables}}$$

Solvency Ratios

$$\text{Debt-to-equity} = \frac{\text{Total debt}}{\text{Total shareholders' equity}}$$

$$\text{Debt-to-capital} = \frac{\text{Total debt}}{\text{Total debt} + \frac{\text{Total shareholders' equity}}{\text{equity}}}$$

$$\text{Debt-to-assets} = \frac{\text{Total debt}}{\text{Total assets}}$$

$$\text{Financial leverage} = \frac{\text{Average total assets}}{\text{Average total equity}}$$

$$\text{Debt-to-EBITDA} = \frac{\text{Total debt}}{\text{EBITDA}}$$

$$\text{Interest coverage} = \frac{\text{EBIT}}{\text{Interest payments}}$$

$$\text{Fixed charge coverage} = \frac{\text{EBIT} + \text{Lease payments}}{\text{Interest payments} + \text{Lease pmts}}$$

Profitability Ratios

$$\text{Net profit margin} = \frac{\text{Net income}}{\text{Revenue}}$$

$$\text{Gross profit margin} = \frac{\text{Gross profit}}{\text{Revenue}}$$

$$\text{Operating profit margin} = \frac{\text{EBIT}}{\text{Revenue}}$$

$$\text{Pretax margin} = \frac{\text{EBT}}{\text{Revenue}}$$

$$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Average total assets}}$$

$$\text{Return on invested capital} = \frac{\text{EBIT}(1 - t)}{\text{ST \& LT debt \& equity}}$$

$$\text{Return on equity (ROE)} = \frac{\text{Net income}}{\text{Average total equity}}$$

DuPont Analysis

$$\text{ROE} = \left(\frac{\text{Net income}}{\text{Assets}} \right) \left(\frac{\text{Assets}}{\text{Book Value of Equity}} \right)$$

$$= (\text{ROA}) \left(\frac{\text{Leverage}}{\text{ratio}} \right)$$

$$= \left(\frac{\text{NI}}{\text{Revenue}} \right) \left(\frac{\text{Revenue}}{\text{Assets}} \right) \left(\frac{\text{Assets}}{\text{Equity}} \right)$$

$$= \left(\frac{\text{Net profit}}{\text{margin}} \right) \left(\frac{\text{Asset}}{\text{turnover}} \right) \left(\frac{\text{Leverage}}{\text{ratio}} \right)$$

$$= \left(\frac{\text{NI}}{\text{EBT}} \right) \left(\frac{\text{EBT}}{\text{EBIT}} \right) \left(\frac{\text{EBIT}}{\text{Revenue}} \right) \left(\frac{\text{Revenue}}{\text{Assets}} \right) \left(\frac{\text{Assets}}{\text{Equity}} \right)$$

$$= \left(\frac{\text{Tax}}{\text{burden}} \right) \left(\frac{\text{Interest}}{\text{burden}} \right) \left(\frac{\text{EBIT}}{\text{margin}} \right) \left(\frac{\text{Asset}}{\text{turnover}} \right) \left(\frac{\text{Financial}}{\text{leverage}} \right)$$

INTRODUCTION TO FINANCIAL STATEMENT MODELING

Behavioral Finance and Analyst Forecasts

- Overconfidence
- Illusion of control
- Conservatism bias
- Representativeness bias
- Confirmation bias

Michael Porter's Five Forces

- Threat of substitutes
- Intensity of rivalry
- Bargaining power of suppliers
- Bargaining power of customers
- Threat of new entrants

CORPORATE ISSUERS

CORPORATE STRUCTURES AND OWNERSHIP

Functions of the Financial System

- Saving
- Borrowing
- Raising Equity Capital
- Managing Risks
- Exchanging Assets
- Information-Motivated Trading

Securities Markets

- *Spot vs. Forward Markets:* Spot market trades are settled within 3 days.
- *Primary vs. Secondary Markets:* Primary market transactions are done directly with the issuer, while secondary market trades take place on organized exchanges.
- *Capital vs. Money Markets:* Money markets are used for securities with maturities of less than one year, while longer-dated securities are traded in capital markets.

Positions

- Long positions:* Benefit from price appreciation
- Short positions:* Benefit from price depreciation



Leveraged Positions

$$\text{Leverage ratio} = \frac{\text{Position}}{\text{Equity}}$$

$$\text{Maximum initial leverage ratio} = \frac{1}{\text{Initial margin}}$$

Maintenance margin: minimum amount of equity required

Margin call is triggered if the equity falls below the maintenance margin. Additional equity will be requested to bring the account balance back to the initial margin.

Execution Instructions (How to fill)

- **Market:** Fill immediately at market price
- **Limit:** Buy below maximum price or sell above minimum price specified in order
- **All-or-nothing:** Cancel order if not fully filled
- **Hidden:** Visible to brokers and exchanges, but invisible to other traders
- **Iceberg:** Only a fraction of order amount is visible

Validity Instructions (When to fill)

- **Day orders:** Cancelled if unfilled at end of day
- **Good-till-cancelled:** No set expiry date
- **Good-on-close:** Filled at end of day
- **Stop-loss:** Sell if prices fall below specified level

Clearing Instructions

Settlement/clearing typically done by brokers for retail trades; brokers or custodians for institutional trades

Primary Market Transactions

- **Initial Public Offerings (IPOs)**
- **Private placements**
- **Shelf registrations:** Part of issue is held back to be sold directly to secondary market investors later
- **Dividend reinvestment plans (DRIPs):** Investors can roll over dividend payments to purchase new shares, possibly at a discount
- **Rights offerings:** Current shareholders gain right to purchase additional shares at below-market price; dilutes value of existing shares

Market Structure

Quote-driven: Investors trade with dealers
Order-driven: Exchanges use order matching rules
Brokered: Trades arranged by brokers
Call markets: Conduct periodic single price auctions, otherwise completely illiquid
Continuous Trading markets: Allow trades whenever market is open, may use call market auction at beginning and/or end of each day

Trade Pricing Rules

Uniform pricing rules: Used by call markets, all trades executed at the price that maximizes total quantity traded
Discriminatory pricing rules: Used by continuous markets, fills most aggressively priced orders first
Derivative pricing rules: Used by crossing networks to trade at midpoint of quotes from other markets

Complete Markets

- Facilitate savings/investment
 - Facilitate lending to creditworthy borrowers
 - Allow risk exposures to be hedged
 - Facilitate exchange of currencies/commodities
- An ideal financial system is *complete* (see above), *operationally efficient* (low transaction costs), and *informationally efficient* (prices reflect all info.)

SECURITY MARKET INDEXES

Price Return over Single Period

$$PR_i = \frac{P_{i1} - P_{i0}}{P_{i0}} \quad PR_i = \sum_{i=1}^N w_i PR_i$$

Total Return over Single Period

$$TR_i = \frac{P_{i1} - P_{i0} + \text{Inc}_i}{P_{i0}} \quad TR_i = \sum_{i=1}^N w_i TR_i$$

Price Return Index over Multiple Periods

$$V_{PRIT} = V_{PRIO}(1 + PR_{I1})(1 + PR_{I2}) \dots (1 + PR_{IT})$$

Total Return Index over Multiple Periods

$$V_{TRIT} = V_{TRIO}(1 + TR_{I1})(1 + TR_{I2}) \dots (1 + TR_{IT})$$

Price-Weighted Indexes

$$w_i^P = \frac{P_i}{\sum_{j=1}^N P_j}$$

- Like buying one share of each stock
- Advantage is simplicity
- Disadvantage is arbitrary weights
- A stock's weight is halved due to a stock split, requiring an adjustment to the divisor

Equally Weighted Indexes

$$w_i^E = \frac{1}{N}$$

- Like investing the same amount in each stock
- Advantage is simplicity
- Disadvantages are that the impact of large companies is underrepresented and frequent rebalancing is required

Capitalization-Weighted Indexes

$$w_i^M = \frac{Q_i P_i}{\sum_{j=1}^N Q_j P_j}$$

- Like holding all stocks in proportion to their market values
- Float adjustment may be used to reflect the number of shares that may be actively traded
- Advantage is that the asset classes' performance is well-represented
- Disadvantage is that returns are heavily driven by large-cap (possibly overvalued) firms

Fundamentally Weighted Indexes

- Built like price-weighted indexes, but using a fundamental measure such as sales or cash flows
- Contrarian effect of rebalancing by selling off top performers and buying underperforming stocks produces a value tilt

Types of Equity Market Indexes

- **Broad market indexes:** Covers one equity market
- **Multi-market indexes:** Covers equity markets in multiple countries
- **Sector indexes:** Important for assessing a manager's performance (selection vs. allocation)
- **Style indexes:** Large/small cap; Value/growth

MARKET EFFICIENCY

Forms of Efficient Market Hypothesis (EMH)

Form	Market Prices Reflect:		
	Past market data	Public info	Private info
Weak	✓		
Semi-strong	✓	✓	
Strong	✓	✓	✓

Implications of EMH

- If weak form holds, investors will not earn abnormal profits from technical analysis
- If markets are semi-strong efficient, investors must have a comparative advantage to earn abnormal profits from fundamental analysis

Market Anomalies

Changes in a security's price that are not attributable to known information

Selected Behavioral Biases

- **Loss aversion:** Disliking losses more than liking equivalent gains
- **Information Cascades:** Those who act first will convey information that influences others
- **Representativeness:** Rely too much on current state when assessing probabilities
- **Mental accounting:** Keep track of gains and losses separately for different investments/goals
- **Conservatism:** Failing to incorporate new information in a timely manner
- **Narrow framing:** Focusing on certain issues in isolation

OVERVIEW OF EQUITY SECURITIES

Common Share Voting Methods

Statutory: One vote per share

Cumulative: Votes can be bundled

Example: 10 board positions

Statutory: 1 share votes for 10 different candidates

Cumulative: 10 votes may go to 1 candidate

Cumulative method advantages small shareholder

Preference Shares

- **Cumulative:** Accrue dividends if payments missed
- **Non-cumulative:** Missed dividends do not accrue, but no common dividends allowed if preferred shareholders do not receive their dividend
- **Participating:** May receive additional dividend if firm is profitable or in the event of liquidation
- **Non-participating:** No compensation beyond dividends and face value in a liquidation



Private Equity Securities

- *Venture capital (VC)*: Start-up, early-stage, or mezzanine financing with IPO as exit strategy
- *Leveraged buyouts (LBO)*: Debt-financed deals to take undervalued listed companies private
- *Private investment in public equity (PIPE)*: Companies can raise new capital quickly, investors can negotiate discounts

Depository Receipts (DRs)

Sponsored DRs: Issued directly by foreign company; Investors receive same voting rights and dividends as other common shareholders

Unsponsored DRs: Foreign company not involved; Depository bank purchases shares, issues DRs, and retains voting rights

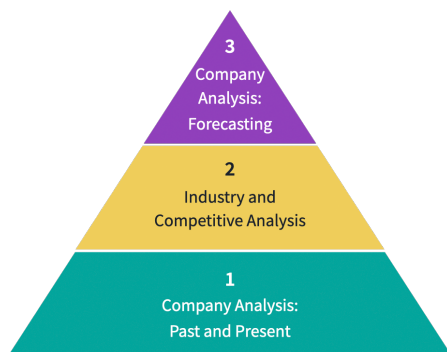
Global DRs: Issued outside company's home country to avoid limits on capital flows; May be denominated in any currency, but USD is common; Cannot be listed on US exchanges, but US investors can purchase them via private placements

American DRs: USD-denominated GDRs that can be traded on US exchanges; Underlying securities, American depository shares, trade in issuer's domestic market

Global Registered Shares: Traded on multiple exchanges, including issuer's domestic market; Denominated in multiple local currencies; Unlike DRs, GRS represent an actual ownership interest

COMPANY ANALYSIS: PAST AND PRESENT

Company Research Reports



Operating Profitability and Working Capital Analysis

Operating income = $[Q \times (P - VC)] - FC$

$$DOL = \frac{\% \Delta \text{Operating income}}{\% \Delta \text{Sales}}$$

$$DFL = \frac{\% \Delta \text{Net income}}{\% \Delta \text{Operating income}}$$

$$DTL = DFL \times DOL$$

INDUSTRY AND COMPETITIVE ANALYSIS

Industry and Competitive Analysis Steps

- (1) Define industry
- (2) Industry survey
- (3) Industry structure
- (4) External influences
- (5) Competitive analysis

External Influences on Industry Growth

- Political influences
- Economic influences
- Social influences
- Technological influences
- Legal influences
- Environmental influences

COMPANY ANALYSIS: FORECASTING

Forecasting Revenues

Top-down approaches

- Growth relative to GDP growth
- Market growth and market share

Bottom-up approaches

- Volume and average sale price
- Product-line or segment revenues
- Capacity-based measures
- Returns- or yield-based measures

EQUITY VALUATION: CONCEPTS AND BASIC TOOLS

Dividend Discount Model (DDM)

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t} = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

Perpetual preferred stock; constant dividend:

$$V_0 = \frac{D_0}{r}$$

Gordon constant growth model (GGM):

$$V_0 = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t} = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

$$g = (\text{Retention rate}) \times \text{ROE} = (1 - D/E) \times \text{ROE}$$

Multistage DDM:

$$V_0 = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

$$P_n = \frac{D_{n+1}}{r - g_L}$$

$$D_{n+1} = D_0(1+g_S)^n(1+g_L)$$

Price Multiples

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r - g}$$

P/B = Price per share/Book value per share

P/CF = Price per share/Cash flow per share

P/S = Price per share/Sales per share

Asset-Based Valuation Models

Useful for companies with natural resource or a large share of current assets/liabilities; Not useful if company has a large share of PP&E/intangibles

Enterprise Value (EV)

$$= MV(\text{Common equity}) + MV(\text{Preferred stock}) + MV(\text{Debt}) - (\text{Cash} + \text{Short term investments})$$

FIXED INCOME

FIXED-INCOME FEATURES, CASH FLOWS, ISSUANCE, AND TRADING

Fixed-Income Cash Flow Structures

- *Bullet Bonds*: Full principal repaid at maturity
- *Fully Amortized Loan*: Equal annuity-like payments contain a mix of interest and principal
- *Partially Amortized Loan*: Some principal is amortized, remainder repaid as a lump sum at maturity
- *Sinking Funds*: Certain percentage of principal retired each year
- *Floating-Rate Notes (FRNs)*: Reference rate + spread
- *Step-Up Bonds*: Coupon rate increases on schedule
- *Sustainability-Linked Bonds*: Coupon rate increases if fails to meet environmental targets
- *Credit-Linked Notes*: Coupon rate increases if issuer is downgraded, reduced if upgraded
- *Payment-In-Kind (PIK) Bonds*: Coupons may be paid with more bonds rather than cash
- *Indexed-Linked Bonds*: Coupons are tied to a specific index
- *Deferred (Split) Coupon Bonds*: No coupons in early years, high coupons in later years

Fixed-Income Contingency Provisions

Callable Bonds

May be recalled by issuer if rates fall

$$V_{\text{Callable bond}} = V_{\text{Non-callable bond}} - V_{\text{Call}}$$

Puttable Bonds

May be sold back to issuer if rates rise

$$V_{\text{Puttable bond}} = V_{\text{Non-puttable bond}} + V_{\text{Put}}$$

Convertible Bonds

Conversion price: Price per share at which bond can be converted into shares

Conversion ratio: Number of common shares each bond can be converted into

Conversion value:

Current share price $\times$ Conversion ratio

Conversion premium:

Convertible bond's price $-$ Conversion value

Warrants: Options to buy equity, lowers debt costs

Convertible Contingent Bonds (CoCos)

- Automatically convert to equity if a condition is met (e.g., capitalization ratio falls)
- Lender does not control if option is exercised
- Primarily issued by financial institutions

Fixed-Income Indexes and Markets

- *Primary bond markets*: Markets in which issuers initially sell bonds to investors to raise capital
- *Secondary bond markets*: Markets in which bonds are subsequently traded among investors; Most trading is OTC rather than on organized exchanges

FIXED-INCOME MARKETS

External Loan Financing

Lines of credit:

- Uncommitted lines of credit
- Committed lines of credit
- Revolving credit agreements

Secured loans and factoring:

- Secured (asset-based) loans
- Factoring

External, Security-Based Financing

- Commercial paper (CP)
- Backup line of credit

Repurchase (Repo) Agreements

$$\text{Repo price} = \frac{\text{Purchase price}}{1 + \text{Repo rate} \left(\frac{\text{Days}}{360} \right)}$$

$$\text{Initial margin} = \frac{\text{Security price}}{\text{Purchase price}}$$

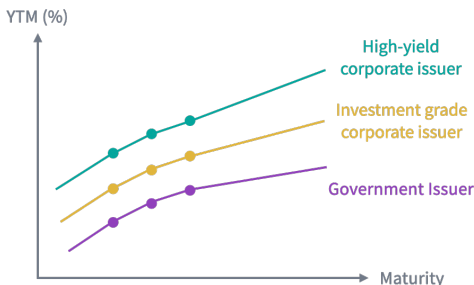
$$\text{Haircut} = \frac{\text{Security price} - \text{Purchase price}}{\text{Security price}}$$

$$\text{Variation margin} = \left(\text{Initial margin} \times \frac{\text{Purchase price}_t}{\text{Purchase price}_0} \right) - \text{Security price}_t$$

Factors Influencing Repo Rates

- Money market interest rates
- Collateral quality
- Repo term
- Collateral uniqueness
- Collateral delivery

Long-Term Corporate Debt



Sovereign Debt

- Issued by national governments, zero default risk
- Bills mature < 1 year; Bonds mature > 1 year
- On-the-run: Most recent issues of a given maturity; more liquid than off-the-run issues

Non-Sovereign Debt

- Municipal bonds (sub-national issuers)
- Quasi-government bonds (gov't-backed agencies)
- Supranational bonds (i.e., IMF, World Bank)

FIXED-INCOME VALUATION: PRICES, YIELDS, INTEREST RATES, AND TERM STRUCTURE

Bond Pricing with a Market Discount Rate

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT + FV}{(1+r)^N}$$

Par	PMT = r	PV = FV
Discount	PMT < r	PV < FV
Premium	PMT > r	PV > FV

Flat Price, Accrued Interest, and Full Price

$$PV^{\text{Full}} = PV^{\text{Flat}} + AI = (PV)(1+r)^{t/T}$$

$$AI = (t/T) \times PMT$$

Relationship Between Bond Prices and Bond Features

- *Inverse effect*: Price moves opposite to yield
- *Convexity effect*: Falling yield has greater price impact than equivalent increase in yield
- *Coupon effect*: Yield changes have greater impact on lower coupon bonds
- *Maturity effect*: Yield changes have greater impact on longer-term bonds (may not apply to low-coupon bonds trading at very deep discounts)
- *Constant-yield price trajectory*: Bonds trading at a discount/premium will be "pulled to par"

Other Yield Measures and Embedded Options

$$\text{Current yield} = \frac{\text{Annual cash coupon payment}}{\text{Flat price}}$$

$$\text{Simple yield} = \frac{\text{Annual cash coupon payment} + \frac{\text{Amortized gain/loss}}{\text{Flat price}}}{\text{Flat price}}$$

Yield-to-call (YTC) = IRR assuming bond is called

Yield-to-worse = min[YTC, YTM]

Yield Spread Measures for Fixed-Rate Bonds

Benchmark spread = YTM - Benchmark yield

G-spread = YTM - Government bond yield

I-spread = YTM - Swap rate

Z-spread

$$PV = \frac{PMT}{(1+z_1+Z)^1} + \frac{PMT}{(1+z_2+Z)^2} + \dots + \frac{PMT + FV}{(1+z_N+Z)^N}$$

(Can only be calculated by trial-and-error)

Option-adjusted spread (OAS)

OAS = Z-spread - Option value (in basis points)

Yield and Spread Measures for FRNs

Quoted margin (QM): Spread paid by FRN

Discount margin (DM): Spread required by market

If QM > DM, FRN will trade above par

FRN pricing formula:

$$= \frac{\frac{(\text{Ref} + \text{QM})(FV)}{m}}{\left(1 + \frac{(\text{Ref} + \text{DM})}{m}\right)^1} + \dots + \frac{\frac{(\text{Ref} + \text{QM})(FV)}{m}}{\left(1 + \frac{(\text{Ref} + \text{DM})}{m}\right)^N} + FV$$

Yield and Spread Measures for Money Market Instrument

Discount Rate (DR) Basis

$$PV = FV \times \left(1 - \frac{\text{Days}}{\text{Year}} \times DR\right)$$

Add-on Rate (AOR) Basis

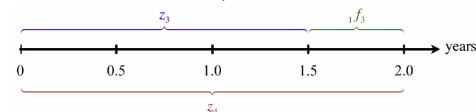
$$PV = FV / \left(1 + \frac{\text{Days}}{\text{Year}} \times AOR\right)$$

Bond Pricing with Spot Rates

$$PV = \frac{PMT}{(1+z_1)^1} + \frac{PMT}{(1+z_2)^2} + \dots + \frac{PMT + FV}{(1+z_N)^N}$$

Forward Rates from Spot Rates

$$(1+z_A)^A \times (1+IFR_{A,B-A})^{B-A} = (1+z_B)^B$$

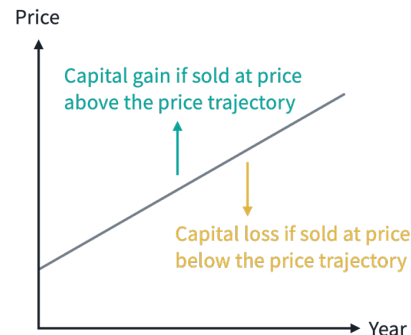


Spot, Par, and Forward Yield Curves

Spot Curve	Par Curve	Forward Curve
Upward sloping	Below spot curve	Above spot curve
Flat	Equal to spot curve	Equal to spot curve
Inverted	Above spot curve	Below spot curve

FIXED-INCOME RISK AND RETURNS

Constant Yield Price Trajectory



Macaulay Duration

$$D_{\text{Mac}} = \left(\frac{1+r}{r} - \frac{1+r+N(c-r)}{c[(1+r)^N - 1] + r} \right) - \frac{t}{T}$$

r: YTM

c: Coupon rate

N: Number of periods to maturity

t: Number of days since last coupon payment

T: Number of days in each coupon period

Duration Gap

Duration gap = D_{Mac} - Investment horizon

- If positive: Price risk > Reinvestment risk
- If negative: Reinvestment risk > Price risk

Yield-Based Duration Measures

Yield duration: Sensitivity to YTM

Measures: Macaulay duration, modified duration, money duration, price value of basis point (PVBP)

Curve duration: Sensitivity to benchmark yields (e.g., effective duration); for bonds with options

Modified Duration

$$\text{ModDur} = \frac{D_{\text{Mac}}}{1 + r}$$

$$\% \Delta PV^{\text{Full}} = -\text{AnnModDur} \times \Delta \text{Yield}$$

$$\text{ApproxModDur} = \frac{(PV_-) - (PV_+)}{2(\Delta \text{Yield})(PV_0)}$$

Money Duration

$$\text{MoneyDur} = \text{AnnModDur} \times PV^{\text{Full}}$$

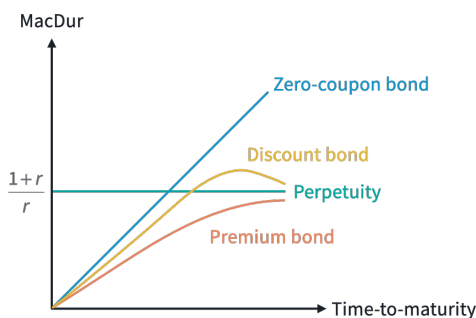
$$\Delta PV^{\text{Full}} \approx -\text{MoneyDur} \times \Delta \text{Yield}$$

Price Value of a Basis Point (PVBP)

$$\text{PVBP} = \frac{(PV_-) - (PV_+)}{2}$$

$$\text{Basis point value (BPV)} = \text{MoneyDur} \times 0.0001$$

Properties of Duration



Bond Convexity and Convexity Adjustment

$$\% \Delta PV^{\text{Full}} = -D_{\text{Mod}} \times \Delta \text{YTM} + \frac{1}{2} (\text{Conv}) (\Delta \text{YTM})^2$$

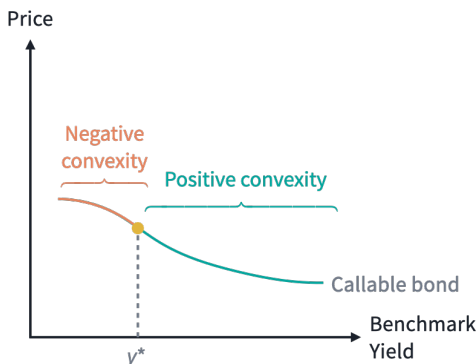
$$\text{Convexity of each cash flow} = \frac{T \times (T + 1) \times w}{\left(1 + \frac{r}{m}\right)^m}$$

$$\text{ApproxCon} = \frac{(PV_-) + (PV_+) - [2 \times (PV_0)]}{(\Delta \text{Yield})^2 (PV_0)}$$

Curve-Based Interest Rate Risk Measures

$$\text{EffDur} = \frac{(PV_-) - (PV_+)}{2(\Delta \text{Curve})(PV_0)}$$

$$\text{EffConv} = \frac{(PV_-) + (PV_+) - 2(PV_0)}{(\Delta \text{Curve})^2 (PV_0)}$$



FIXED-INCOME: CREDIT RISK AND ANALYSIS

C's of Credit Analysis

Bottom-up:

- Capacity
- Capital
- Collateral
- Covenants
- Character

Top-down:

- Conditions
- Country
- Currency

Measuring Credit Risk

- Default risk: Probability of default

- Loss severity: Loss given default

$$E[\text{Loss}] = \text{Pr}(\text{Default}) \times \text{Loss severity}$$

$$\text{Loss severity} = 1 - \text{Recovery rate}$$

Credit Rating Agencies and Credit Ratings

Credit migration risk: Possibility of downgrade

Credit ratings:

- *Investment grade:* Baa3/BBB- and above
- *Non-investment grade:* Ba1/BB+ and below

Seniority Ranking

- First Lien Loan – Senior Secured
- Second Lien Loan – Secured
- Senior Unsecured
- Senior Subordinated
- Subordinated
- Junior Subordinated

Pari passu: All creditors in the same ranking, regardless of maturity, have the same priority

FIXED-INCOME SECURITIZATION: ASSET-BACKED AND MORTGAGE-BACKED SECURITIES

Securitization Example

- Firm sells equipment on credit
- Firm creates bankruptcy-remote SPV
- SPV issues debt to purchase loans from firm
- SPV creates securities backed by loans
- Investors purchase securities from SPV
- SPV collects loan payments from firm's customers
- SPV distributes cash flows to investors

Parties to a Securitization and Their Roles

- *Seller/Depositor:* Originates loans (assets)
- *Issuer:* Special purpose vehicle (SPV) established to create asset-backed securities (ABS)
- *Servicer:* Collects payments on underlying loans

Covered Bonds

- Dual recourse against the issuing financial institution and the cover pool
- One bond class per cover pool
- Issuer must replace non-performing asset with performing asset

Non-Mortgage ABS

- *Amortizing:* E.g., auto loan ABS
- *Non-amortizing:* E.g., credit card receivable ABS

Collateralized Debt Obligations (CDO)

Securities backed by pool of debt obligations, such as corporate bonds, leveraged bank loans, or credit default swap on securities

Mortgage Loans

Agency RMBS: Issued by government agencies; must have conforming loans

Non-agency RMBS: Issued by private companies and may have non-conforming loans

Features:

- Prepayment: penalty, no penalty
- Foreclosure: non-recourse, recourse

Residential Mortgage-Backed Securities (RMBS)

- *Pass-through rate:* MBS coupon rate
- Unlike pass-through securities, CMOs have tranches to redistribute cash flows and risks
- *Sequential-pay CMOs* have principal and prepayments paid to the tranches sequentially
- *Prepayment risk:* Contraction (faster-than-expected); extension (slower-than-expected)

CMO Tranches

- Z-tranches
- Principal-only (PO) tranches
- Interest-only (IO) tranches
- Floating-rate tranches
- Residual tranches
- Planned amortization class (PAC)

Commercial Mortgage-Backed Securities (CMBS)

$$\text{DSC} = \frac{\text{Net annual operating income (NOI)}}{\text{Debt service}}$$

$$\text{NOI} = \left(\text{Rental income} \times \frac{\text{Cash operating expense}}{\text{Reserves}} \right) - \text{Replacement reserves}$$

$$\text{Debt service} = \frac{\text{Annual interest payments}}{\text{Debt service}} + \frac{\text{Annual principal repayment}}{\text{Debt service}}$$

DERIVATIVES

DERIVATIVE INSTRUMENT AND DERIVATIVE MARKET FEATURES

Derivative Underlyings

- Equities
- Fixed-income instruments and interest rates
- Currencies
- Commodities
- Credit
- Other (e.g., weather, crypto, longevity risk)

Derivatives Markets

	OTC Market	ETD Market
Liquidity	Lower	Higher
Trading costs	Higher	Lower
Transparency	Less	Greater
Standardization	Lower	Higher
Flexibility/customization	Higher	Lower
Counterparty credit risk	Higher	Lower

FORWARD COMMITMENT AND CONTINGENT CLAIM FEATURES AND INSTRUMENTS

Types of Derivatives

Forward commitments: Obligation to trade on a specified date at a previously agreed price

Contingent claims: Trade may or may not occur depending on market conditions

Forward Commitments	Contingent Claims
Forward contract	Options
Futures contracts	Credit derivatives
Swaps	

DERIVATIVE BENEFITS, RISKS, AND ISSUER AND INVESTOR USES

Derivative Benefits and Risks

Benefits:

- Risk allocation, transfer, and management
- Information discovery
- Operational advantages
- Market efficiency

Risks:

- Potential for speculative use
- Lack of transparency
- Basis risk
- Liquidity risk
- Counterparty credit risk
- Destabilization and systemic risk

Derivative Benefits and Risks

Issuers use derivatives to perform:

- Cash flow hedge
- Fair value hedge
- Net investment hedge

Investors use derivatives to hedge their risk exposures and take speculative positions

ARBITRAGE, REPLICATION, AND THE COST OF CARRY IN PRICING DERIVATIVES

No-Arbitrage Forward Price with Benefits/Income and Costs

Annual compounding:

$$F_0(T) = [S_0 + PV_0(C) - PV_0(I)](1+r)^T$$

Continuous compounding:

$$F_0(T) = S_0 e^{(r+c-i)T}$$

No-Arbitrage Currency Forward Price

$$F_{0,f/d}(T) = S_{0,f/d} e^{(r_f - r_d)T}$$

PRICING AND VALUATION OF FORWARD CONTACTS

Pricing and Valuation of Forward Commitments

Valuation at initiation:

$$F_0(T) = S_0(1+r)^T$$

Valuation at maturity:

$$V_T(T) = S_T - F_0(T)$$

- Long party:

$$V_T(T) = -[S_T - F_0(T)]$$

Valuation during the life of the contract:

$$V_t(T) = S_t - F_0(T)(1+r)^{-(T-t)}$$

Pricing and Valuation of Currency Forward Contracts

Valuation at initiation:

$$F_{0,f/d}(T) = S_{0,f/d} e^{(r_f - r_d)T}$$

Valuation during the life of the contract:

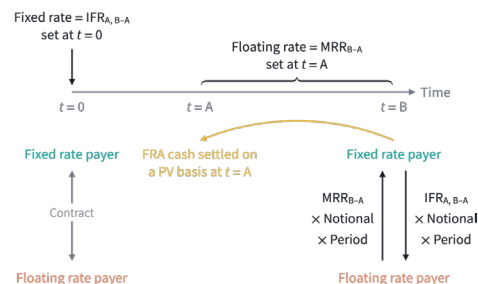
$$V_t(T) = S_{t,f/d} - F_{0,f/d}(T) e^{-(r_f - r_d)(T-t)}$$

Pricing and Valuation of Interest Rate Forward Contracts

Implied forward rate:

$$IFR_{A,B-A} = \left[\frac{(1+z_B)^B}{(1+z_A)^A} \right]^{1/(B-A)} - 1$$

Forward rate agreements:



Net payment received by the FRA buyer:

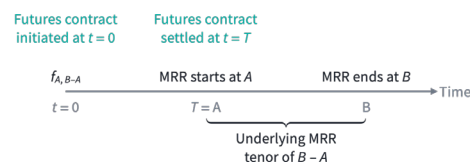
$$(MRR - \text{Fixed IFR}) \times \text{Notional principal} \times \text{Period}$$

PRICING AND VALUATION OF FUTURES CONTACTS

Pricing of Futures and Forward Contracts

Price of interest rate futures contract:

$$f_{A,B-A} = 100 - (100 \times MRR_{A,B-A})$$



Type	Gains from Rising MRR	Gains from Falling MRR
Interest rate futures	Short futures contract	Long futures contract
Forward rate agreement	Long FRA: Fixed-rate payer	Short FRA: Floating-rate payer

Futures Contract Value

$$= \text{Notional Principal} \times \left[1 + \left(\frac{MRR}{M} \right) \right]$$

Futures Contract BPV

$$= \text{Notional Principal} \times 0.0001 \times \text{Period}$$

Forward Price vs. Futures Price

The relationship between forward prices and futures prices depends on the correlation between futures prices and interest rates:

Correlation	Relationship
None	Futures price = Forward price
Positive	Futures price > Forward price
Negative	Futures price < Forward price

PRICING AND VALUATION OF INTEREST RATES AND OTHER SWAPS

Swap Values and Prices

Periodic settlements:

Periodic Settlement Value

$$= (MRR - \text{Swap rate}) \times \text{Notional} \times \text{Period}$$

PRICING AND VALUATION OF OPTIONS

Option Moneyness

Option Moneyness	Call	Put
In-the-money	$S_t > X$	$S_t < X$
At-the-money	$S_t = X$	$S_t = X$
Out-of-the-money	$S_t < X$	$S_t > X$

Option Values

$$c_T = \max[0, S_T - X]$$

$$p_T = \max[0, X - S_T]$$

$$c_t = \max[0, S_t - X/(1+r)^{T-t}] + \text{Time value}$$

$$p_t = \max[0, X/(1+r)^{T-t} - S_t] + \text{Time value}$$

Arbitrage and Replication for Options

Call option:

$$\text{Lower bound } (c_t) = \max[0, S_t - X/(1+r)^{T-t}]$$

$$\text{Upper bound } (c_t) = S_t$$

Put option:

$$\text{Lower bound } (p_t) = \max[0, X/(1+r)^{T-t} - S_t]$$

$$\text{Upper bound } (p_t) = X$$

Factors Impacting Option Values

Increase in	Call	Put
Value of underlying	↑	↓
Exercise price	↓	↑
Time to expiration	↑	↑*
Risk-free rate	↑	↓
Volatility of underlying	↑	↑
Payments on underlying	↓	↑
Cost of carry	↑	↓

*Except for some deep-in-the-money put options

OPTION REPLICATION USING PUT-CALL PARITY

Put-Call Parity

$$c_0 + X/(1+r)^T = s_0 + p_0$$

Fiduciary call = Protective put

Put-Call-Forward Parity

$$c_0 + X/(1+r)^T = F_0(T)/(1+r)^T + p_0$$

Fiduciary call = Protective put w. forward contract

Put-Call Parity Applications: Firm Value

$$V_0 + p_0 = c_0 + PV(D)$$

VALUING A DERIVATIVE USING A ONE-PERIOD BINOMIAL MODEL

Binomial Valuation

Hedging portfolio:

$$V_0 = hS_0 - c_0$$

Hedge ratio:

$$h = \frac{c_1^u - c_1^d}{S_1^u - S_1^d}$$

Risk Neutrality

Risk-neutral probabilities:

$$\pi = \frac{1+r-d}{u-d}$$

Risk-neutral probability-weighted call price:

$$c_0 = \frac{\pi c_1^u + (1-\pi)c_1^d}{1+r}$$

ALTERNATIVE INVESTMENTS

ALTERNATIVE INVESTMENT FEATURES, METHODS, AND STRUCTURES

Alternative Investments: Features & Categories

- Narrow manager specialization
- Low correlation with traditional investments
- Large capital outlays
- Long investment horizons
- Illiquidity
- Alternatives to direct investment
- Incentive-based compensation arrangements
- Performance appraisal challenges

Alternative Investment Methods

- *Fund Investing*: Indirect investing
- *Co-Investing*: Hybrid between direct investing and indirect investing
- *Direct investing*: Without the use of intermediary

Compensation Structures

- *Soft hurdle rate*: Incentive fee applies to entire return if hurdle rate is cleared

$$r < r_h \rightarrow \text{Performance fee} = 0$$

$$r \geq r_h \rightarrow \text{Performance fee} = p \times r$$

- *Hard hurdle rate*: Incentive fee is only paid on return in excess of hurdle rate

$$\text{Performance fee} = \max[0, p(r - r_h)]$$

- *Catch-up clause*: Allows the GP to receive 100% of the return in excess of the hurdle rate until the GP catches up with their cumulative performance fee

$$\text{Performance fee} = \max[0, r_{cu} + p(r - r_h - r_{cu})]$$

- *High-water mark clause*: Reflects the highest value used to calculate an incentive fee

- *Clawback provision*: Allows the LPs to get back incentive fees that have been paid if gains are subsequently reversed

- *Waterfall*: Distribution method that defines the order of allocations to the LPs and GPs

ALTERNATIVE INVESTMENT PERFORMANCE AND RETURNS

Multiple of Invested Capital

$$MOIC = \frac{\left(\frac{\text{Realized value}}{\text{of investment}} \right) + \left(\frac{\text{Unrealized value}}{\text{of investment}} \right)}{\text{Total amount of investment}}$$

Alternative Investment Returns

Restrictions on withdrawals:

- Redemption fees
- Notice periods
- Lock periods
- Liquidity gates

Alternative Investment Return Calculations

R_{GP} (Net with hurdle rate)

$$= (r_m \times P_1) + \max\{0, p[P_1(1 - r_m) - P_0(1 + r_h)]\}$$

INVESTMENTS IN PRIVATE CAPITAL: EQUITY AND DEBT

Private Equity: Leveraged Buyouts

- *Management buyouts*: Current management team is involved in the acquisition
- *Management buy-ins*: Current management team is being replaced by the acquiring team

Private Equity: Venture Capital

- *Formative-stage financing*: Angel investing, seed-stage financing, early-stage financing
- *Later-stage financing*: After commercial production and sales have begun but before IPO
- *Mezzanine-stage financing*: Prepare to go public

Private Equity Exit Strategies

- Trade sale
- Public listing (IPO, direct listing, SPAC)
- Recapitalization
- Secondary sale
- Liquidation

Private Debt

- *Venture debt*: To complement existing equity financing of start-up or early-stage companies
- *Direct lending*: Direct capital in the form of senior and secured loans
- *Mezzanine debt*: Debt subordinated to senior secured debt but senior to equity
- *Distressed debt*: Funding provided to mature companies facing financial distress

REAL ESTATE AND INFRASTRUCTURE

Real Estate Investment Structures

	Debt	Equity
Private	Mortgages, Construction lending	Direct ownership, Real estate funds, Private REITs
Public	MBS, CMOs, Mortgage REITs	Shares in RE corps., REITs

Real Estate Fund Structures

Infinite-life open-end funds:

- Core real estate
- Core-plus real estate

Finite-life closed-end funds:

- Value-add
- Opportunistic

Benefits of Real Estate Investments

- Stable, predictable income
- Price appreciation
- Inflation hedging
- Portfolio diversification
- Tax benefits

Infrastructure Investment Features

- New (greenfield) or existing (brownfield) assets
- Economic (roads) or social (healthcare facilities)
- Direct ownership or indirect (via MLP or ETF)

NATURAL RESOURCES

Timberland and Farmland

Sources of return:

- Biological growth
- Prices of timber/crops
- Land price changes

Commodity Investment Forms

Futures price:

$$S_0(1+r) + \text{Cost of carry} - \text{Convenience yield}$$

$$F_0(T) = S_0 e^{(r+c-t)T}$$

	Contango	Backwardation
Price curve slope	Upward	Downward
Convenience yield	Low	High



HEDGE FUNDS

Hedge Fund Strategies

- **Equity hedge:** Long and short positions in equity and equity derivative securities; Bottom-up
- **Event-driven:** Seek to profit from short-term events (e.g., Mergers); Bottom-up
- **Relative value:** Seek to profit from pricing discrepancies between related securities
- **Opportunistic:** Emphasize top-down approach to identifying global economic trends

Hedge Fund Features

- Low legal and regulatory restrictions
- Large investment universe
- Use of derivatives and short positions
- Leverage
- Aggressive strategies with concentrated positions
- Limits on liquidity
- High fees

INTRODUCTION TO DIGITAL ASSETS

Distributed Ledger Technology

- Tokenization:** Represents ownership rights to physical assets on a distributed ledger
- Non-fungible tokens (NFTs)
 - Security tokens
 - Utility tokens
 - Governance tokens

Digital Asset Investment Forms

Direct	Indirect
Centralized exchanges	Cryptocurrency coin trusts
Decentralized exchanges	Cryptocurrency futures contracts
	Cryptocurrency exchange-traded funds
	Cryptocurrency stocks
	Hedge funds

PORTFOLIO MANAGEMENT

PORTFOLIO RISK AND RETURN: PART I

Risk Aversion

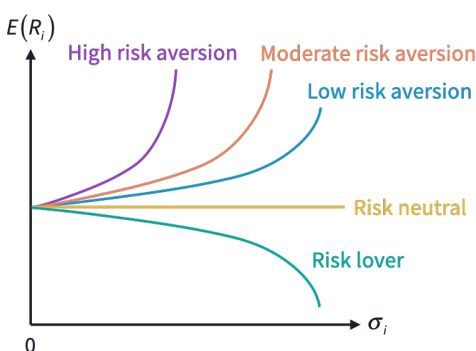
Combination of ability and willingness to take risk
Risk averse: Requires a premium to take more risk
Risk neutral: Only concerned with expected return, indifferent to level of risk
Risk seeking: Will pay a premium to take more risk

Utility Function

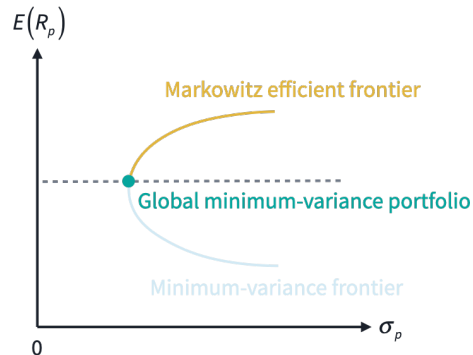
$$U = E(r) - \frac{1}{2} A \sigma^2$$

A is the degree of risk aversion, it is >0 for risk-averse, 0 for risk-neutral, and <0 for risk-seeking

Indifference Curves



Minimum-Variance Portfolios

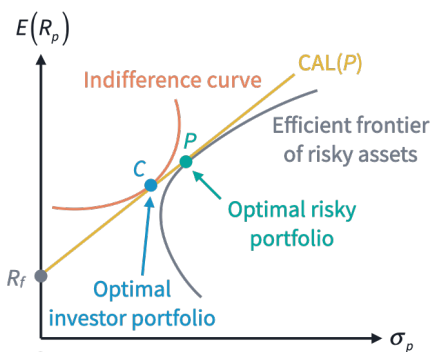


Capital Allocation Line (CAL)

Line representing possible combinations of risk-free assets and optimal risky asset portfolio

$$E[R_p] = R_f + \left(\frac{E[R_i] - R_f}{\sigma_i} \right) \sigma_p$$

Investor's Optimal Portfolio



PORTFOLIO RISK AND RETURN: PART II

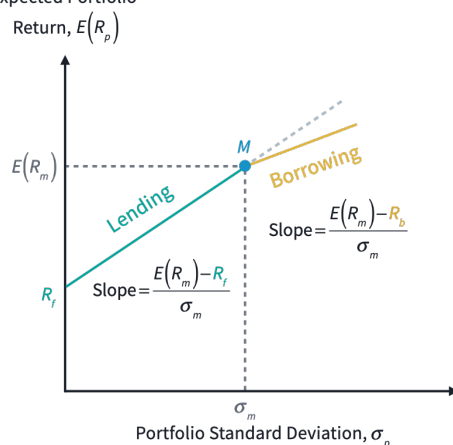
Capital Market Line (CML)

CAL with risky portfolio being market portfolio

$$E[R_p] = R_f + \left(\frac{E[R_m] - R_f}{\sigma_m} \right) \sigma_p$$

Borrowing vs. Lending

Expected Portfolio Return, $E(R_p)$



Beta

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \frac{\rho_{i,m} \sigma_i}{\sigma_m}$$

Systematic risk = Non-diversifiable (market) risk

Nonsystematic risk = Diversifiable risk

Total risk = Systematic risk + Nonsystematic risk

Capital Asset Pricing Model (CAPM)

Assumptions:

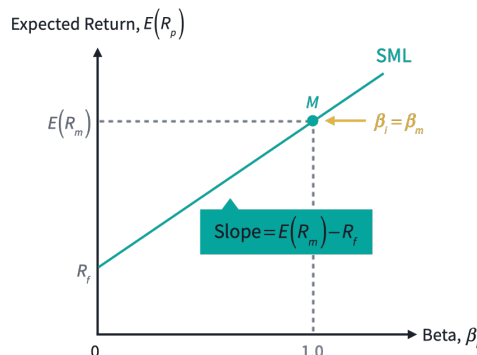
- Investors are risk-averse, utility-maximizing, rational individuals
 - Markets are frictionless
 - All investors plan for same single holding period
 - Investors have homogeneous expectations
 - Investments are infinitely divisible
 - Investors are price takers
- $$E[R_p] = R_f + \beta_i [E[R_m] - R_f]$$

Limitations:

- Single-factor: Only accounts for systematic risk
- Single-period: Does not consider multiple periods
- Inclusion of assets that are not investable, such as human capital and assets in closed economies

Security Market Line (SML)

Graphical representation of CAPM:



Identifying Mispriced Stocks

Stocks that plot above the SML are underpriced; stocks that plot below the SML are overpriced

Ratios

Total risk	Sharpe ratio	$\frac{R_p - R_f}{\sigma_p}$
	M-squared	$(R_p - R_f) \frac{\sigma_m}{\sigma_p} + R_f$
Systematic risk	Treynor ratio	$\frac{R_p - R_f}{\beta_p}$
	Jensen's alpha	$R_p - [R_f + \beta_p(R_m - R_f)]$

PORTFOLIO MANAGEMENT: AN OVERVIEW

Portfolio Management Process

Planning: List objectives and constraints in IPS

Execution: Asset allocation, security analysis, portfolio construction

Feedback: Monitoring and rebalancing, performance measurement and reporting

Institutional Investor Clients

- *DB pension plans:* Younger beneficiaries increase time horizon and risk tolerance
- *Endowments/Foundations:* Generally longer time horizon, low liquidity needs, high risk tolerance
- *Banks:* Short time horizon, high liquidity need, very low risk tolerance
- *Insurers:* Short time horizon (longer for Life than P&C), high liquidity needs, low risk tolerance
- *Investment companies:* Time horizon and risk tolerance vary by mandate, liquidity needs are usually high due to potential redemptions
- *Sovereign Wealth Funds:* Vary by mandate

Robo-Advisors

- Cater to underserved segments, "mass affluent"
- Lower fees compared to traditional managers
- Relatively low barriers to entry

Mutual Funds

- *Open-end:* Accept new investors after launch
- *Closed-end:* No new shares created after launch, may trade at a premium/discount to NAV
- *No-Load:* No investing/redemption fees, funds charge a percentage of NAV

Exchange Traded Funds (ETFs)

- Mutual funds only trade at the end of each day, ETFs can be traded at any time during the day
- Investors can sell ETFs short or buy on margin
- ETFs do not trade at discount/premium to NAV
- ETFs distribute dividends to investors, mutual funds reinvest dividends
- ETFs have lower minimum investment levels

BASICS OF PORTFOLIO PLANNING AND CONSTRUCTION

Investment Policy Statements (IPS)

Investment objectives: Risk/return objectives

Constraints: Liquidity, time horizon, tax concerns, legal and regulatory factors, unique circumstances

Asset Allocation

Strategic asset allocation: Set of exposures to IPS-permissible asset classes in weights that are consistent with the client's long-term objectives

Tactical asset allocation: Deliberate deviations from policy weights based on forecasts of asset class returns over the near term

THE BEHAVIORAL BIASES OF INDIVIDUALS

Cognitive Errors

- *Conservatism bias:* People fail to incorporate new information that conflicts with their opinions
- *Confirmation bias:* People seek "evidence" that confirms their prior beliefs
- *Representativeness bias:* People inappropriately classify new information based on past similar situations
- *Illusion of control bias:* People overestimate their ability to control or predict events
- *Hindsight bias:* People believe past events would have been predictable
- *Anchoring and adjustment bias:* People rely too much on initial information in their estimation
- *Mental accounting bias:* People put money in separate mental buckets
- *Framing bias:* People answer the same question differently based on how it is framed
- *Availability bias:* People assume outcomes that are easier to remember are more likely

Emotional Biases

- *Loss-aversion bias:* People strongly prefer avoiding losses more than achieving gains
- *Overconfidence bias:* People overestimate their own abilities
- *Self-control bias:* People lack self-discipline to make decisions based on their long-term goals
- *Status quo bias:* People are more inclined to do nothing rather than make changes
- *Endowment bias:* People value an asset more when they hold the rights to it
- *Regret-aversion bias:* People avoid making decisions that could potentially turn out badly

Market Anomalies

Factors that cause anomalies misclassifications:

- Inappropriate asset pricing model
- Statistical issues due to small samples
- Temporary disequilibria

AN INTRODUCTION TO RISK MANAGEMENT

Risk Management

Risk management framework:

- Risk governance
- Risk identification and measurement
- Risk infrastructure
- Defined policies and processes
- Risk monitoring, mitigation, and management
- Communications
- Strategic analysis or integration

Risk tolerance: Which risks are acceptable and how much risk should be taken

Risk budgeting: How the risks should be taken

Financial risks: Arise from financial market activities (e.g., market, credit, liquidity risk)

Non-financial risks: Arise from within entity or from external (e.g., operational, legal, regulatory, political, model, tail risk)

Risk measures: Standard deviation, beta, duration, delta, gamma, VaR, CVaR, etc.

Risk modification: By prevention and avoidance, transfer (insurance), or shifting (derivatives)

ETHICAL AND PROFESSIONAL STANDARDS

I(A) Knowledge of the Law

Obey strictest applicable law. Disassociate immediately from any illegal or unethical activity.

I(B) Independence and Objectivity

Do not offer or accept gifts that might impair independence and objectivity. Gifts from clients may be permissible.

I(C) Misrepresentation

Cite sources. Do not plagiarize or omit important information. Act quickly to correct any errors.

I(D) Misconduct

Does not apply to personal behavior unless it reflects poorly on the investment profession.

II(A) Material Nonpublic Information

Do not act or cause others to act on material nonpublic information. Seek public dissemination.

II(B) Market Manipulation

Do not take any actions that distort prices or trading volume. Market making and legitimate trading strategies are allowed.

III(A) Loyalty, Prudence, and Care

Place clients' interest above yours. Disclose policies on proxy voting and soft commissions.

III(B) Fair Dealing

Treat all clients fairly. Treat non-immediate family like other clients. Communicate investment recommendations and changes simultaneously.

III(C) Suitability

Use a regularly updated IPS during investment decisions. Evaluate decisions in a portfolio context.

III(D) Performance Presentation

Performance data should be fair, accurate, and complete. Do not promise returns for risky assets.

III(E) Preservation of Confidentiality

Keep all client information confidential unless: client is involved in illegal activity, you are legally required, or you have the client's permission.



IV(A) Loyalty

Get permission before taking outside work (even unpaid) that competes with employer. Abide by non-compete agreement (if applicable) and do not take employer's property.

IV(B) Additional Compensation Arrangements

Obtain written permission from all parties before receiving any compensation for outside work.

IV(C) Responsibilities of Supervisors

Supervisors must adequately train and monitor subordinates. Responsibilities may be delegated.

V(A) Diligence and Reasonable Basis

Exercise diligence and thoroughness. Support actions with research and investigation.

V(B) Communication with Clients and Prospective Clients

Make appropriate disclosures. Distinguish between fact and opinion in analysis and recommendations.

V(C) Record Retention

Maintain records to support recommendations and decisions. 7-year retention period recommended.

VI(A) Disclosure of Conflicts

Disclose any matters that may impair independence and objectivity, prominently and in plain language.

VI(B) Priority of Transactions

Execute clients' transactions before accounts in which you have a beneficial interest.

VI(C) Referral Fees

Disclose referral fees to clients and employer, including non-monetary arrangements.

VII(A) Conduct as Participants in CFA Institute Program

Do not share confidential exam details. Expressing opinions about CFAI policies is permissible.

VII(B) Reference to CFA Institute, the CFA Designation, and the CFA Program

Do not misrepresent the meaning of CFA Institute membership, designation, or candidacy.

BA II PLUS CALCULATOR TIPS

Basic Operations

2ND: Access secondary functions (in yellow)

ENTER: Send value to a variable

2ND + **ENTER**: Toggle between options

↑ **↓**: Navigate between variables/options

STO + **0 - 9**: Store current value into memory

RCL + **0 - 9**: Recall value from memory

Time Value of Money (TVM)

For annuity, loan, and bond calculations

N: Number of periods

1/Y: Effective interest rate per period (in %)

PV: Present value

PMT: Payment/coupon amount

FV: Future value/redemption value

CPT + **one of the above**: Solve for unknown

2ND + **BGN**: Toggle between ordinary annuity and annuity due

2ND + **CLR TVM**: Clear TVM worksheet

Note:

- Always clear the TVM worksheet before starting a new calculation
- For bonds, PMT and FV should have the same sign, and opposite signs to PV

Cash Flow Worksheet (**CF**, **NPV**, **IRR**)

For non-level payments

Input (**CF**)

CF0: Initial cash flow

C01: 1st distinct cash flow after initial cash flow

F01: Frequency of C01

C0n: nth distinct cash flow

F0n: Frequency of C0n

Note:

- Always clear the CF worksheet before starting a new calculation
- The use of F0n is optional. You can leave them as 1 and input repeating cash flows multiple times. If you do so, C01 will be the cash flow at time 1, C02 will be the cash flow at time 2, and so on.

Output (**NPV**, **IRR**)

I: Effective interest rate per period (in %)

NPV + **CPT**: Solve for net present value

IRR + **CPT**: Solve for internal rate of return